

Weak-Instrument Inference and First-Stage Concentration in Double Machine Learning for the Partially Linear IV Model

Airui Meng*

July, 2026

Abstract

Double machine learning (DML) for the partially linear instrumental-variables (PLIV) model controls for high-dimensional confounders with flexible learners. Yet flexible first-stage learning can leave the residualized instrument weak and collapse the structural estimate toward zero behind a *small* standard error, as documented by Çetin (2026). We show this collapse is a loss of residualized first-stage *concentration*, not of identification: any consistent first stage targets the same structural parameter, and under a correctly specified treatment nuisance the prediction-optimal first stage in fact *maximizes* the population concentration (Proposition 3), so correctly applied flexible learning imposes no inherent absorption cost. The small standard error is scale-invariant in the residualized instrument and does not signal a precisely estimated zero; the inferential failure is the distributional non-pivotality of the t -ratio. We therefore make a cross-fitted Anderson–Rubin set and effective- F screening the primary reporting layer. A known-truth Monte Carlo verifies that the Anderson–Rubin coverage stays near nominal across the regimes our applications occupy, while Wald coverage deviates far more. A strength-as-treatment design spans a randomized, a high-dimensional, a weak, and an over-identified cell. Across these cells the realized losses trace to two diagnosable sources: finite-sample overfit dilution, and treatment-nuisance and conditioning effects. Under randomization, where the true first stage is constant, neither source operates. An identification-aware first-stage selector that breaks predictive-loss ties toward concentration reduces finite-sample bias at moderate strength and tail risk in the weakest regime, at no cost to consistency or coverage.

Keywords: double machine learning; partially linear instrumental-variables model; weak instruments; first-stage concentration; Anderson–Rubin; cross-fitting; identification-robust inference; first-stage nuisance selection.

JEL classification: C14, C21, C26, C55.

1. Introduction

Double machine learning has made it routine to use flexible, high-dimensional learners for the nuisance functions of a causal model while retaining valid inference on a low-dimensional parameter. For the partially linear instrumental-variables (PLIV) model, Çetin (2026) provides a leakage-safe nested cross-validation protocol that selects nuisance learners without contaminating the held-out fold, and establishes $\sqrt{n}$ -asymptotic normality under strong identification. The protocol’s appeal is

*Columbia University. Email: am6817@tc.columbia.edu. This paper builds on and extends the partially linear IV setting of Çetin (2026). All numerical results are reproduced by the replication materials accompanying the paper.

that the analyst can let the data choose how to control for confounders, including the first-stage map from controls to the instrument, without committing in advance to a parametric specification.

This paper builds on double machine learning, in which a Neyman-orthogonal moment combined with cross-fitting delivers $\sqrt{n}$ -inference on a low-dimensional parameter while the nuisance functions are estimated by flexible learners (Chernozhukov et al. 2018). For the partially linear IV model the orthogonal score residualizes outcome, treatment, and instrument on the controls; Çetin’s (2026) nested protocol supplies the leakage-safe learner selection within this framework. The guarantees in this line presume strong identification, formalized as a residualized concentration bounded away from zero, and report Wald standard errors. They are therefore silent on what a flexible first stage does to that concentration once the assumption is stressed, which is the regime we study; we treat the concentration as the object of interest and pair the orthogonal estimator with inference that remains valid when it is low.

The inference we adopt comes from the weak-instrument literature, which constructs tests whose size is controlled regardless of instrument strength. The Anderson and Rubin (1949) statistic and its refinements (the Kleibergen 2002 score test and the Moreira 2003 conditional likelihood ratio) invert into identification-robust confidence sets, while the Montiel Olea and Pflueger (2013) effective- F screens first-stage strength against worst-case bias and the tF procedure of Lee, McCrary, Moreira, and Porter (2022) adjusts the Wald interval (Andrews, Stock, and Sun 2019 review the area). This machinery was developed for fixed instruments in classical two-stage least squares. Its behavior on a cross-fitted, generated instrument is not established: the residualized instrument is itself an estimated object, carrying generated-regressor heteroskedasticity and second-order nuisance-error products, one of them scaled by the tested τ . The nearest antecedent is Belloni, Chen, Chernozhukov, and Hansen (2012), who pair lasso-selected optimal instruments with a weak-identification-robust sup-score test; their estimator uses the lasso-selected instruments in the structural equation, their sup-score test screens the full instrument set against the structural error without selection, and neither operates on a moment in which all three PLIV nuisances are cross-fitted away. We execute the Anderson–Rubin set and the Montiel Olea–Pflueger effective- F screen on the cross-fitted PLIV moment and verify their coverage by simulation rather than assume it transfers; the over-identified procedures (Kleibergen’s score test, Moreira’s CLR) we run on the over-identified Angrist–Krueger cell (§4.9), where the design calls for them. A closely related strand develops identification-robust inference under many and many-weak instruments, including jackknife and cross-fit corrections to the first-stage bias of two-stage least squares (Hansen, Hausman, and Newey 2008; Chao and Swanson 2005; Kolesár 2013; Mikusheva and Sun 2022), which govern the over-identified and cluster-jackknife cases here.

Our high-dimensional cell draws on the examiner-design literature, in which the leniency of a quasi-randomly assigned decision-maker instruments for a treatment (Dobbie, Goldin, and Yang 2018; Chyn, Frandsen, and Leslie 2025). Because such designs assign many examiners in clusters, leave-one-out and cluster-jackknife estimators correct the many-instrument bias of two-stage least squares (Angrist, Imbens, and Krueger 1999; Frandsen, Leslie, and McIntyre 2025). Angrist and Frandsen (2022) bring machine-learning first stages, including random forests, to this design class and document the biases that machine-generated instruments can carry. We use the cluster-jackknife leniency instrument, validate our implementation against the published estimates, and embed the design in the concentration framework as the cell where a genuinely flexible first stage makes over-fitting of the instrument an economically real concern; our increment over that account is the dilution-versus-absorption diagnosis and the residualized effective- F accounting.

A parallel literature estimates causal effects with flexible instrumental-variables methods that target heterogeneous or structural objects, such as the instrumental forest of Athey, Tibshirani, and Wager (2019) and the Deep IV estimator of Hartford et al. (2017), and studies the implicit weighting

of the linear IV estimand under flexible controls (Blandhol, Bonney, Mogstad, and Torgovitsky 2026). These methods target different estimands than the scalar partially linear effect, so a like-for-like comparison first requires testing for homogeneity. We treat that comparison as a separate exercise and keep the present analysis focused on the concentration and inference of the scalar estimator.

Relative to these strands, our contribution is distinct on three fronts. Relative to the DML line, we treat the residualized concentration itself as the object of interest rather than as an assumption: we show that under a correctly specified treatment nuisance the prediction-optimal first stage *maximizes* it (Proposition 3), and we replace the Wald default with identification-robust inference precisely where the concentration is low. Relative to the weak-instrument literature, we transport the Anderson–Rubin set and the effective- F screen to a *cross-fitted, generated* instrument and verify their coverage there rather than assuming it, including under clustering and survey weights. And relative to the examiner-design and flexible-IV literatures, we use those designs not as applications in their own right but as the arms of a strength-as-treatment contrast that separates an estimator pathology from conditional irrelevance. The existing weak-instrument evidence cannot make that separation.

This flexibility imposes a less obvious cost on the instrument. When a flexible learner is used for the first-stage nuisance $\pi_0(W) = \mathbb{E}[Z \mid W]$, the residualized instrument can become weak, and the estimated effect can collapse toward zero with a standard error that is small rather than large. Çetin (2026) documents exactly this pattern in his CollegeDistance return-to-schooling application: regularized-linear residualization leaves the estimate near its two-stage least-squares value, whereas random-forest nuisance learners (the first stage among them) drive the estimate to approximately zero with a small Wald standard error. A near-zero point estimate beside a small standard error invites the reading that the return is a precise zero. We argue that this reading is wrong, and that correcting it reorganizes both the inference and the empirical design.

Three facts about this collapse are already established, and we credit them to Çetin (2026). He documents the sensitivity and proposes the mechanism reading, namely that an overly flexible first stage may absorb a large share of the identifying variation in the instrument (his Section 5.1); he explains the small standard errors as limited residualized instrument variation rather than a scaling artifact (his Table 2); and he names the residualized concentration quantity that governs precision (his Remark 3), while maintaining strong identification throughout. He also flags weak-instrument-robust inference as the needed remedy but places it out of scope, reporting Wald standard errors. Two problems remain. First, where the residualized concentration is low, the Wald interval is not a valid statement of precision, so the collapse is reported with an inference tool that the same analysis shows to be invalid. Second, the available evidence cannot separate an estimator pathology from a design that was already weak after conditioning, because every instrument in the schooling application is conditionally near-irrelevant, which makes a first-stage estimation artifact and conditional irrelevance observationally equivalent.

We take the concentration view seriously and build the analysis around it. Our starting point is that the choice of a consistent first stage cannot change the estimand: any consistent estimator of π_0 targets the same structural parameter, so the collapse is a loss of residualized first-stage concentration, not of identification, and the small standard error is scale-invariant in the residualized instrument. Indeed, under a correctly specified treatment nuisance the prediction-optimal first stage *maximizes* the population concentration (Proposition 3), so the loss is never an inherent cost of flexible learning. It is finite-sample dilution or a nuisance-specification effect. The inferential failure is distributional (the non-pivotality of the t -ratio under weak concentration), which is why weak-instrument-robust inference must be primary rather than supplementary. On this basis we make five contributions.

- **Weak-instrument-robust inference, executed.** We make effective- F screening and a cross-fitted Anderson–Rubin set the default reporting layer, executing the remedy Çetin flags as future work; its coverage is certified in the known-truth Monte Carlo of the fourth contribution rather than assumed (Sections 3 and 4).
- **A strength-as-treatment design.** We vary the residualized first-stage strength across a randomized cell, a high-dimensional first-stage cell, and a weak cell, which separates an estimator pathology from a design that is already weak after conditioning, a separation the existing weak-instrument evidence cannot make (Section 4).
- **The concentration view formalized, with a new optimality result.** We supply the theoretical core the sensitivity was missing: under a correctly specified treatment nuisance the prediction-optimal first stage *maximizes* the population residualized concentration (Proposition 3, our central theoretical result), so flexibility per se carries no absorption cost. Two framing results organize the rest: the estimand is invariant to the choice of consistent first stage (Proposition 1), and the inferential failure is distributional rather than a matter of standard-error scale (Proposition 2). These results formalize the sensitivity Çetin documents and the classical weak-instrument distortion, so that a near-zero estimate with a small standard error is never read as a precise zero (Section 2).
- **A known-truth Monte Carlo in the weak regime.** We trace concentration to coverage with a known structural parameter: the coverage certificate of §4.2, including its clustered and survey-weighted arms, certifies that the cross-fitted set keeps coverage close to nominal, and far closer than Wald, across the regimes the data occupy; and varying the treatment-nuisance specification distinguishes the misspecified- r interaction from any genuine population cost and finds none (Proposition 3), in a way the observational cells alone cannot (Section 4).
- **An identification-aware first-stage selector.** We select the first stage to maximize residualized concentration among the learners tied on predictive loss, subject to the orthogonality rate condition, which modestly reduces the estimator’s finite-sample bias at moderate strength and its tail risk in the weakest regime by breaking predictive-loss ties toward a more concentrated first stage. The gain persists under correctly specified nuisances and vanishes asymptotically (Proposition 3), without altering coverage (Section 4). Çetin’s (2026) restricted- π sensitivity analysis, which limits the first-stage library to guard against unstable learners, is the precursor; our increment is a continuous, rate-condition-constrained tie-break rather than a binary full-versus-restricted library switch.

The empirical design sharpens the mechanism rather than merely confirming it. Correctly specified, flexible learning does not erode the residualized first stage (Proposition 3); the realized losses across the cells trace to two diagnosable sources, and to neither under randomization. In the high-dimensional examiner cell a flexible first stage cannot predict the orthogonalized instrument out of fold but overfits it in sample, diluting concentration through cross-fitted noise, a finite-sample estimation pathology. The known-truth simulation shows that the remaining apparent “absorption” is not a population cost of flexible π but an interaction with the treatment nuisance: with a correctly specified r the flexible first stage attains the *higher* concentration, and the effective- F drop appears only when $\hat{r}$ underfits $\mathbb{E}[D \mid W]$. The weak schooling cell, once its geographic fold leakage is removed, is a weak/ambiguous cell whose flexible-learner movement is an overfitting artifact; the Card control-richness ladder additionally exhibits the conditioning-set route, in which richer controls lower the conditional relevance through $r(W)$. In the randomized cell the first stage is constant by construction, so neither route operates and the estimate is stable. An identification-robust Anderson–Rubin set is the valid interval throughout. The conclusion is that a near-zero estimate under flexible learning is a weak-instrument, finite-sample, and nuisance-specification phenomenon,

identifiable from standard diagnostics, rather than a defect of using machine learning.

The remainder of the paper proceeds as follows. Section 2 sets out the PLIV model and identification. Section 3 presents estimation and inference. Section 4 contains the empirical applications. Section 5 discusses limitations, and Section 6 concludes. Appendices A and B collect the proofs and the full result tables.

2. Model and Identification

2.1 Setup

We study the partially linear instrumental-variables (PLIV) model

$$Y = \tau_0 D + g_0(W) + \zeta, \quad \mathbb{E}[\zeta \mid Z, W] = 0,$$

where D is a scalar endogenous regressor, Z is an instrument of dimension d_z , W is a vector of controls, and τ_0 is the scalar structural parameter of interest. Estimation rests on the residualized, Neyman-orthogonal PLIV score

$$\psi(\tau, \eta) = (Z - \pi(W))[(Y - \mu(W)) - \tau(D - r(W))],$$

with nuisance functions $\mu_0(W) = \mathbb{E}[Y \mid W]$, $r_0(W) = \mathbb{E}[D \mid W]$, and $\pi_0(W) = \mathbb{E}[Z \mid W]$. We write residualized variables as $\tilde{Y} = Y - \mu_0(W)$, $\tilde{D} = D - r_0(W)$, and $\tilde{Z} = Z - \pi_0(W)$.

We take as given the leakage-safe nested cross-validation protocol of Çetin (2026): within each outer cross-fitting fold, learners and hyperparameters for (μ, r, π) are selected by inner cross-validation on the outer-training sample only, refit on the full outer-training fold, and evaluated on the held-out fold. Under a product-rate condition on the nuisance estimators and strong identification (Çetin’s Assumption 2, that the residualized concentration is bounded away from zero), his Theorem 2 delivers $\sqrt{n}$ -asymptotic normality of the residualized-IV estimator. Our contribution begins where that strong-identification assumption is relaxed.

2.2 The estimand does not depend on the choice of consistent first stage

The first observation organizes the rest of the paper. The role of π in the score is to residualize the instrument, not to identify τ_0 .

Proposition 1 (Estimand invariance and the role of the first stage). *Suppose the PLIV model holds with $\mathbb{E}[\zeta \mid Z, W] = 0$, that the outcome and treatment nuisances are consistent ($\hat{\mu} \rightarrow \mu_0$, $\hat{r} \rightarrow r_0$), and that $\mathbb{E}[\tilde{Z}\tilde{D}] \neq 0$. Then for **any** first-stage estimator $\hat{\pi}$ converging to a square-integrable, W -measurable limit $\pi_*(W)$ (not necessarily π_0), the cross-fitted residualized-IV estimator is consistent for the **same** τ_0 . The choice of first stage affects only the residualized concentration C , and through it precision, not the estimand or consistency (Appendix A.1 records the scope of the precision claim).*

The argument is that the residualized instrument is a valid multiplier for *any* first stage. Because $\mathbb{E}[\zeta \mid Z, W] = 0$, for any W -measurable π_* we have $\mathbb{E}[(Z - \pi_*(W))\zeta] = \mathbb{E}[Z\mathbb{E}[\zeta \mid Z, W]] - \mathbb{E}[\pi_*(W)\mathbb{E}[\zeta \mid Z, W]] = 0$, so the moment $\mathbb{E}[\psi(\tau_0, \eta)] = 0$ holds at the truth whatever (square-integrable) limit the first stage takes. Flexible learners for π therefore cannot change identification, the estimand, or consistency; they change only the residualized first-stage strength. Two consequences organize the rest of the paper. First, a collapse of $\hat{\tau}$ toward zero under flexible learning is never evidence that “the true effect is zero” or that “identification was destroyed.” Second, a linear first stage is consistent for τ_0 *even when π_0 is nonlinear*, so when two learners disagree

neither is the biased one on first-stage grounds; the disagreement is a statement about concentration (and, if the outcome or treatment nuisance is misspecified, about those nuisances) rather than about the instrument.

2.3 Concentration governs precision, and the failure mode is distributional

The first stage, the treatment nuisance, and the conditioning set together govern the residualized first-stage strength. Define the per-observation residualized concentration

$$C = g'_{ZD} \Omega^{-1} g_{ZD}, \quad g_{ZD} = \mathbb{E}[\tilde{Z}\tilde{D}], \quad \Omega = \mathbb{E}[\tilde{Z}\tilde{Z}'],$$

and the concentration parameter $\mu^2 = nC$. The asymptotic variance of the residualized-IV estimator scales as $\text{Var}(\hat{\tau}) \propto 1/(nC) = 1/\mu^2$, exactly so under conditional homoskedasticity of the score (Appendix A.1); under heteroskedasticity C ranks weak-identification strength, so a reduction in C is a loss of identification strength at fixed n . We show below (Proposition 3) that under a correctly specified treatment nuisance C is in fact *maximized* at the true first stage π_0 , so correctly applied flexible learning does not reduce it in population; §2.4 then catalogs the finite-sample and specification routes by which the *realized* concentration can fall.

Proposition 2 (Distributional, not scale, failure). *In the just-identified case $d_z = 1$, the Wald variance is $V = \Sigma/g_{ZD}^2$ with $\Sigma = \mathbb{E}[\tilde{Z}^2 u^2]$ and $u = \tilde{Y} - \tau_0 \tilde{D}$. (i) This V is **invariant to the scale of $\tilde{Z}$** : a small reported standard error therefore carries no evidence about instrument strength on its own, and along the sequence in (ii) the standard error remains $O_p(1)$: bounded, not vanishing. (ii) Under the Staiger and Stock (1997) local-to-zero (weak-instrument) concentration sequence (nC bounded as $n \rightarrow \infty$), however, the studentized statistic $(\hat{\tau} - \tau_0)/\hat{\text{se}}$ is **non-pivotal** whenever the endogeneity correlation $\rho \neq 0$; its distribution departs from the standard normal, and the nominal $1 - \alpha$ Wald interval fails to attain $1 - \alpha$ coverage despite the small standard error (Appendix A.2).*

Proposition 2 corrects a tempting but wrong reading of the schooling collapse, in which a near-zero estimate with a tiny standard error is taken as a precise zero. The standard error is small because g_{ZD} is small (limited residualized instrument variation), not because the variance has blown up, and the inferential failure is the non-pivotal sampling distribution, not the magnitude of the standard error. The practical consequence, developed in §3, is that weak-instrument-robust inference must be the primary reporting layer whenever C is low, rather than an optional diagnostic.

We do not discover this pattern: as credited in §1, Çetin (2026) documents the sensitivity, reads the small standard errors as limited residualized instrument variation rather than a scaling artifact, and names the concentration quantity $g'_{ZD} \Omega^{-1} g_{ZD}$, all under maintained strong identification. What we add is its anatomy: Propositions 1–2 formalize the sensitivity as a concentration-and-distribution statement, and Proposition 3 (§2.4) shows that the natural reading of it (flexible π *absorbing* the instrument) does not survive under correctly specified nuisances, so the realized loss is finite-sample dilution or a treatment-nuisance specification effect rather than an inherent cost of flexible learning.

2.4 The anatomy of concentration loss

Proposition 2 says low C breaks Wald inference; it does not say what lowers C . The organizing result is that, correctly specified, flexible first-stage learning does *not* lower it in population.

Proposition 3 (Concentration is maximized at the true first stage). *With the treatment nuisance correctly specified, $r = r_0 = \mathbb{E}[D | W]$, the residualized concentration $C(\pi) = g_{ZD}(\pi)' \Omega(\pi)^{-1} g_{ZD}(\pi)$ satisfies $C(\pi) \leq C(\pi_0)$ for every square-integrable W -measurable π , with equality iff $\pi = \pi_0$ when*

$d_z = 1$, and for $d_z > 1$ iff the perturbation $\pi_0 - \pi$ is concentration-neutral ($M \Omega(\pi_0)^{-1} g_{ZD} = 0$, with $M = \mathbb{E}[(\pi_0 - \pi)(\pi_0 - \pi)']$; Appendix A.3). The prediction-optimal first stage π_0 is therefore concentration-optimal in population: a consistent flexible learner that recovers a nonlinear π_0 attains the **maximum** concentration and cannot lower it.

The proof is short (Appendix A.3), and it closes only the population channel: realized, finite-sample concentration can still fall, through the finite-sample and specification routes listed below. Under $r = r_0$ the relevance $g_{ZD}(\pi) = \mathbb{E}[\tilde{Z}_0 \tilde{D}_0]$ is independent of π : the W -predictable part of Z carries *zero* residualized covariance with $\tilde{D}$, because $\tilde{D}$'s W -structure is already removed by r_0 . Meanwhile $\Omega(\pi) = \Omega(\pi_0) + \mathbb{E}[(\pi_0 - \pi)(\pi_0 - \pi)']$ is minimized at π_0 . Recovering π_0 shrinks the residualized instrument toward its signal, raising C .

It follows that a concentration *loss* under flexible learning is never an inherent population cost of using a flexible π . Such a loss has two diagnosable sources, separated by the out-of-fold predictive R^2 of $\hat{\pi}$ on Z ($R_{\text{of}}^2(\hat{\pi}; Z)$) and the residualized-variance ratio $\text{Var}(\tilde{Z})/\text{Var}(Z)$.

Diagnostic characterization (routes by which the *realized* concentration falls). • **Overfit dilution**: a finite-sample first-stage-*estimation* pathology. $\hat{\pi}$ has no out-of-fold predictive gain, $R_{\text{of}}^2(\hat{\pi}; Z) \leq 0$, yet a large in-sample overfit gap; cross-fitting converts the overfit into idiosyncratic noise in the held-out residual, so $\text{Var}(\tilde{Z})/\text{Var}(Z) > 1$ and g_{ZD} is diluted. The dilution vanishes as $n \rightarrow \infty$ and is removed by a restricted or cross-fit-stabilized $\hat{\pi}$.

- **Treatment-nuisance and conditioning effects**: operating through r and W , not π . The realized C depends on the conditioning set and on $\hat{r}$: (i) a richer set of controls W genuinely lowers the *conditional* relevance $\text{Cov}(\tilde{Z}, \tilde{D})$; this is a modeling choice, read off C (the Card ladder, §4.7), and it operates through $r(W)$ rather than π ; (ii) an *under-specified* $\hat{r} \neq \mathbb{E}[D | W]$ breaks the π -invariance of g_{ZD} that underlies Proposition 3 and can manufacture an apparent instrument “absorption” (a flexible $\hat{\pi}$ then *appears* to lower C), which a correctly specified flexible $\hat{r}$ removes (§4.8). Under randomization of Z given W , π_0 is constant, no learner predicts Z in or out of fold, $\tilde{Z} \approx Z$, and C is preserved: neither route operates.

The signatures therefore read as follows. $R_{\text{of}}^2(\hat{\pi}; Z) > 0$ with $\text{Var}(\tilde{Z})/\text{Var}(Z) < 1$ means the flexible learner is *correctly* recovering W -predictable structure in Z (a nonlinear π_0); under correct nuisances this *raises* C (Proposition 3), so this signature alone is not a loss. A loss appearing alongside it signals the misspecified- r interaction. $R_{\text{of}}^2(\hat{\pi}; Z) \leq 0$ with $\text{Var}(\tilde{Z})/\text{Var}(Z) > 1$ is overfit dilution. The examiner cell instantiates overfit dilution, the Card ladder the conditioning/ r -route, and the known-truth Monte Carlo the misspecified- r interaction; the OHIE cell is the randomized control where neither route operates. No cell exhibits a population absorption cost of flexible π , and by Proposition 3 none can.

3. Estimation and Inference

3.1 Cross-fitted residualized estimation

For each dataset we cast the problem into (Y, D, Z, W) , estimate $(\hat{\mu}, \hat{r}, \hat{\pi})$ by leakage-safe cross-fitting (the nested protocol of Çetin 2026, run here in its cluster-aware form), form the held-out residuals $(\tilde{Y}, \tilde{D}, \tilde{Z})$, and solve the empirical analogue of $\mathbb{E}[\psi] = 0$. For $d_z > 1$ we use the optimally weighted residualized GMM estimator

$$\hat{\tau} = \frac{\hat{g}'_{ZD} \hat{\Omega}^{-1} \hat{g}_{ZY}}{\hat{g}'_{ZD} \hat{\Omega}^{-1} \hat{g}_{ZD}}, \quad \hat{g}_{ZY} = \frac{1}{n} \sum_i \tilde{Z}_i \tilde{Y}_i,$$

with the standard sandwich variance. Three implementation choices matter for the empirical cells and are stated here because they affect validity, not merely efficiency. First, cross-fitting folds are **cluster-aware**: whole clusters (judge shifts, households) are assigned to a single fold, so within-cluster information cannot leak across the train/held-out split. Second, all reported standard errors, first-stage F statistics, and identification-robust statistics use a cluster-robust variance at the relevant level. Third, the outcome, treatment, and first-stage nuisances are estimated with **separate** learners, each selected by its own out-of-fold prediction loss, rather than a single learner applied to all three; this isolates the first-stage learner under study and prevents a misspecified outcome or treatment nuisance from being confounded with it (a coupling that, in §4.6, manufactured a spurious disagreement between linear and random-forest first stages).

3.2 Identification-robust inference is the primary layer

Çetin (2026) flags weak-instrument-robust inference as needed but out of scope (his Remark 3 and Conclusion) and reports Wald standard errors throughout. Proposition 2 implies that, in any low-concentration cell, those Wald intervals are not valid precision statements. We therefore execute identification-robust inference as the default reporting layer, not a supplementary diagnostic.

The procedure has two stages. We first **screen** with the Montiel Olea–Pflueger (2013) effective- F computed on the cross-fitted residualized moment, read against their worst-case-bias tables rather than the rule of ten; because those critical values are calibrated for a fixed instrument, we use the effective- F as a strength screen and rely on the AR coverage simulation (§4.2), not the effective- F threshold, to certify inference. We then report, for every estimator row, a cross-fitted **Anderson–Rubin (AR)** confidence set obtained by inverting the AR statistic over a grid of τ , alongside the Wald interval, and we flag rows where the two diverge. For $d_z = 1$ (the case of every empirical cell here) the AR, Kleibergen-LM, and Moreira-CLR sets coincide, so we report the AR set directly. An over-identified design would additionally invoke the Kleibergen-LM and Moreira-CLR sets and a Hansen–Sargan over-identification test (Sargan 1958; Hansen 1982); the over-identified empirical cell that exercises these is the Angrist–Krueger quarter-of-birth design (§4.9). (With multiple endogenous regressors, a case that does not arise here, the first-stage screen would instead be the conditional F of Sanderson and Windmeijer 2016.) The AR variance is computed with a heteroskedasticity- or cluster-robust per- τ form, because the residualized moment carries generated-regressor heteroskedasticity, and the set is reported as a set, including unbounded or empty outcomes, following Davidson and MacKinnon (2014).

One caveat is internal to the cross-fitted construction and motivates §4.2. The PLIV moment $\mathbb{E}[\tilde{Z}(\tilde{Y} - \tau\tilde{D})]$ is first-order Neyman-orthogonal in all three nuisance directions at *every* τ , because the multiplier $\tilde{Z}$ is conditional-mean-zero given W and, at the true (μ_0, r_0) , so is the bracket $\tilde{Y} - \tau\tilde{D}$. What survives is second order: the products of the first-stage estimation error with the outcome- and treatment-nuisance errors enter the moment, the treatment product scaled by the tested τ . The product-rate condition controls these terms at any fixed τ , but the far reaches of an inverted confidence set lean hardest on that control, and the generated instrument contributes estimation noise of its own. A naive $\chi^2_{d_z}$ reference for the cross-fitted AR statistic is therefore not guaranteed to be uniformly valid; a locally robust correction in the sense of Chernozhukov, Escanciano, Ichimura, Newey, and Robins (2022) would suppress the second-order terms at every τ , and we leave that construction to future work (§5). Rather than assume validity, we verify it by simulation, in the coverage certificate of §4.2.

4. Empirical Applications

4.1 Design: residualized first-stage strength as the treatment

A portfolio of weak instruments cannot separate an estimator pathology from a design in which the instrument is already conditionally irrelevant, because “flexible $\hat{\pi}$ absorbs Z ” and “ Z was conditionally irrelevant” are then observationally equivalent. We instead vary the residualized first-stage strength as the object of interest, across a three-cell contrast that spans its support: a randomized cell where π_0 is constant by construction, a high-dimensional cell where the first stage is a genuinely flexible function of W , and a weak cell where the instrument is known to be weak after conditioning. The weak cell (CollegeDistance, §4.6) doubles as a stress test of the cross-fitting protocol: its region-determined outcome and original coupled-nuisance analysis make it a leakage-and-misspecification illustration rather than a clean mechanism cell, while the Card cell (§4.7) supplies the clean schooling estimand and a within-instrument concentration ladder. Each cell carries a sharp prediction under §2.4. The cross-cell synthesis is Figure 1. The over-identified Angrist–Krueger cell (§4.9) is reported separately, as it tests the over-identified weak-instrument inference of §3.2 and §4.2 rather than the strength-as-treatment contrast.

The cells deliberately span the identification spectrum, and we read each against the inference layer of §3. The Oregon lottery is the strong-identification case (effective- $F \approx 2100$): the Wald and Anderson–Rubin intervals coincide, and the point estimate carries its usual interpretation. Card’s lean specification is moderate but genuinely identified (effective- F 14–17), and the examiner cell sits between regimes: moderately identified under a ridge first stage (25.5) but pushed toward the weak side by flexible-learner dilution (15.0). CollegeDistance (8.8–11.0), the upper rungs of the Card control ladder ($7 \rightarrow 3$), and the $d_z = 30$ Angrist–Krueger block are the weak-identification cases: there, following §3.2, the Anderson–Rubin set is the only interval we interpret, and a near-zero estimate beside a small Wald standard error is read as low concentration rather than as a precise zero (Proposition 2). The contrast also shows where the leakage-safe protocol earns its keep: in the strong-identification cells the cluster-aware discipline merely confirms the design (OHIE’s effective- F is flat across learners), while in the weak cells it is decisive, reversing the CollegeDistance diagnosis by removing the spurious absorption that shuffled folds manufacture (R_{oof}^2 0.78 $\rightarrow$ 0.02, §4.6). The weaker the identification, the more the conclusions depend on the cross-fitting discipline and on the identification-robust layer.

Table 1 reports descriptive statistics for each cell, Table 2 the residualized first-stage strength by nuisance learner, and Table 3 the main estimates (OLS, 2SLS, and DML) with their identification-robust intervals. The (quasi-)randomization the OHIE and examiner designs rely on is established in the source studies (the OHIE lottery and the within-court-by-time assignment of examiners) rather than recomputed here.

Table 1: Descriptive statistics and covariate balance

Cell	N	clusters	Y mean	D mean	Z mean (SD)	balance
OHIE	23,441	20,747	1.95	0.279	0.50	by design
Examiner	91,282	1,831	0.585	0.327	0.03 (0.03)	by design
CollegeDistance	4,739	433	2.24	13.81	1.80	—
Card	3,010	robust	6.26	13.26	0.68	—

Notes. Y : OHIE primary-care visits, examiner conviction, CollegeDistance log state wage, Card individual log wage; D : ever-Medicaid, pretrial release, years of education; Z : lottery, cluster-jackknife leniency, distance, growing up near a four-year college. Clusters: households, judge-by-shift assignments, state proxies (Card uses heteroskedasticity-robust standard errors). "Balance: by design" indicates that the (quasi-)randomization is established in the source study rather than recomputed here: OHIE by lottery within household-size strata (Finkelstein et al. 2012), examiner leniency by quasi-random assignment within court-by-time with the covariate-balance test reported by Chyn, Frandsen, and Leslie (2025). CollegeDistance and Card rely on conditional-on-observables identification and carry no design-based balance test.

Table 2: Residualized first-stage strength (effective- F) by nuisance learner

Cell	ridge	RF	GB
OHIE	2100	2038	2112
Card	16.5	14.5	17.1
Examiner	25.5	15.0	23.6
CollegeDistance	11.0	8.8	9.1

Notes. Effective- F on the cross-fitted, **cluster-aware, decoupled-nuisance** residualized first stage. OHIE uses the survey-weighted responder primary specification (§4.4); the other three cells are unweighted. Only the examiner cell shows a clear dilution drop under a flexible learner; OHIE and Card stay flat, and CollegeDistance is uniformly weak (its earlier flexible-learner "collapse" was fold leakage, §4.6). *Sample construction.* OHIE: 74,922 lottery records merged to the state-program and twelve-month survey files, restricted to the twelve-month survey responders with non-missing outcome, treatment, ever-Medicaid, household size, survey weight, and household identifier, leaving 23,441 weighted responders in 20,747 households. Examiner: 94,210 Miami-Dade cases restricted to judges with at least 200 cases, leaving 91,282 cases in 1,831 shifts. CollegeDistance: 4,739 complete cases with positive wage.

Table 3: Main results: estimates and identification-robust inference

Cell	Estimator	$\hat{\tau}$	Wald SE	eff- F	AR 95% set	N
OHIE	OLS	1.196	0.063	—	—	23,441
	2SLS	1.034	0.173	2089	—	23,441
	DML (ridge)	1.046	0.173	2100	[0.71, 1.39]	23,441
	DML (RF)	1.044	0.175	2038	[0.70, 1.39]	23,441
Examiner	OLS	−0.158	0.005	—	—	91,282
	2SLS	−0.497	0.189	24.4	—	91,282
	DML (ridge)	−0.455	0.142	25.5	[−0.91, −0.06]	91,282
	DML (RF)	−0.406	0.174	15.0	[−1.04, 0.11]	91,282
CollegeDistance	OLS	−0.001	0.001	—	—	4,739
	2SLS	0.052	0.020	27.6	—	4,739
	DML (ridge)	−0.005	0.004	11.0	[−0.013, 0.002]	4,739
	DML (RF)	−0.011	0.005	8.8	[−0.024, −0.004]	4,739
Card	OLS	0.075	0.004	—	—	3,010
	2SLS	0.132	0.054	13.5	—	3,010
	DML (ridge)	0.132	0.053	16.5	[0.03, 0.27]	3,010
	DML (RF)	0.139	0.055	14.5	[0.04, 0.29]	3,010

Notes. $\hat{\tau}$ with cluster-robust (OHIE on household, examiner on judge-by-shift) or heteroskedasticity-robust (CollegeDistance, Card) Wald standard errors. The effective- F is the Montiel Olea–Pflueger screen on the cross-fitted residualized first stage; the **AR 95% set** is the cross-fitted Anderson–Rubin interval, the primary weak-instrument-robust interval (§3.2), and is the column to read when the effective- F is small. The DML (ridge) row uses a genuinely cross-validated ridge first stage (λ selected over eight decades); it coincides with the unpenalized linear benchmark on every cell (effective- F within 0.2 at the moderate-strength cells and within 0.01% at OHIE’s ≈ 2100 ; point estimates within 0.006), so separate linear rows are omitted as redundant. The gradient-boosted estimates are reported with the full estimator panel in Appendix B (Table B.1). The over-identified Angrist–Krueger cell is reported separately in §4.9. Reading across cells: where the residualized first stage is strong (OHIE) or moderate but real (Card) the estimate is stable across learners with a bounded AR set; in the weak conditional design (CollegeDistance) the flexible first stage pulls $\hat{\tau}$ toward zero behind a small Wald standard error, while the AR set remains the honest interval.

4.2 A coverage certificate for the cross-fitted AR set

The estimates of Table 3 take the cross-fitted Anderson–Rubin set as the primary interval. We first establish, in a known-truth Monte Carlo, that this set keeps coverage close to nominal across the regimes the empirical cells occupy, and far closer than the Wald interval, reporting Monte-Carlo standard errors with each coverage estimate. The data-generating process is the PLIV model with known (τ_0, π_0) ; a tuning parameter sweeps the residualized concentration so that the realized effective- F ranges from about 1 to about 500, with $d_z \in \{1, 3, 5, 10, 30\}$ and both oracle and cross-fitted nuisances. The primary strength sweep adapts the Monte-Carlo design of Çetin (2026) to the weak-identification regime; the adversarial stress panel and the over-identified, nonlinear-nuisance, clustered, and survey-weighted arms are new.

The certificate has six components. A primary sweep varies the first-stage strength under oracle, estimated-linear, and **estimated-flexible (random-forest)** first stages, the last reproducing the flexible-first-stage regime of the empirical cells. An adversarial stress panel then probes six additional corners not covered by the sweep (including tiny $n = 400$; near-total residualization with π explaining 85% of Z ; $d_z = 10$; the very-weak-and-flexible worst case). A third, over-identified layer matches the Angrist–Krueger cell (§4.9) directly, sweeping the concentration at $d_z = 3$ and $d_z = 30$ and adding the Kleibergen–LM and Moreira–CLR coverage and the Hansen–Sargan J -test size that the over-identified case requires. A fourth arm replaces the linear structural functions g_0 and r_0 with nonlinear ones recovered by random forests (all three nuisances (μ, r, π) then carry estimation

error) and finds the cross-fitted AR coverage at or above nominal (0.976 and 0.968 at effective- F of about 2.9 and 9.8, if anything conservative), confirming that the certificate’s validity does not rest on linear structural nuisances. A fifth arm makes the data *clustered*: shared cluster effects enter the structural error and a cluster-level component enters the instrument, as in the examiner-shift and OHIE-household cells. This arm confirms that the cluster-robust cross-fitted AR retains nominal coverage (0.953 and 0.958 at effective- F of about 5.8 and 8.5, over 200 clusters of size 25 and 10) exactly where a naive heteroskedasticity-only variance under-covers badly (0.51 and 0.71); the cluster-robust inference the clustered cells rely on is therefore itself coverage-backed. A sixth arm exercises the survey-weighted path of the OHIE primary specification: a survey stratum carries a heterogeneous effect, so that the survey-population effect τ_w differs from the unweighted-sample effect τ_u , as in OHIE, where weighting moves the estimand. The survey-weighted cross-fitted AR covers τ_w at nominal (0.959 and 0.960 at effective- F of about 44 and 157), whereas a naive unweighted AR, centered on the wrong estimand τ_u , covers it only 0.56 and 0.07 of the time, the gap widening as the instrument strengthens; survey weighting is therefore necessary, and the weighted inference is itself coverage-backed.

The pattern is the following (Figure 2; the full panel is Table B.2), with Monte-Carlo standard errors reported alongside each coverage estimate. Across the oracle, estimated-linear, and over-identified regimes and the eleven adversarial stress points, the cross-fitted AR coverage ranged from 0.943 to 0.962 (target 0.95; the stress panel alone bottomed out at 0.947, and the $d_z = 5$ over-identified point at 0.943). In the regime that most resembles the applications (an estimated flexible random-forest first stage on a nonlinear design), AR coverage was 0.945 and 0.943 at two calibrated design points (median effective- F 9.9 and 33.3, chosen to bracket the flexible-learner strengths the cells occupy; 1,500 replications, Monte-Carlo standard error ≈ 0.006), modestly below the nominal 0.95 by about one Monte-Carlo standard error, reflecting the finite-sample noise that a flexible, generated instrument injects into the moment and its variance at the null; we report this regime explicitly rather than excluding it, and note that the Wald interval over-covers here (0.96) while badly mislocating the point estimate. Wald, by contrast, failed where the theory predicts: in the over-identified design (with oracle, fixed-linear first-stage nuisances; i.e., the classical over-identified weak-IV Wald bias, not a cross-fitting artifact) its coverage fell to 0.91 at $d_z = 5$ and to 0.79 at $d_z = 10$, and in the just-identified design its point estimate was severely biased at weak instruments (a mean bias of about -1.6 , inflated by heavy-tail draws in this very weak regime, at effective- $F \approx 2.1$ under an estimated-linear first stage) even where the inflated standard error mechanically covered. At the over-identified operating points the identification-robust sets hold where Wald does not: across the Angrist–Krueger regimes ($d_z = 3$ and $d_z = 30$ at effective- F from about four to thirty-seven, 1,200 replications each), the cross-fitted AR, Kleibergen-LM, and Moreira-CLR coverage rates all lie between 0.953 and 0.962 (Monte-Carlo standard error ≈ 0.006), and the Hansen–Sargan J test has size 0.044 at both $d_z = 3$ (2 d.f.) and $d_z = 30$ (29 d.f.), so the identification-robust sets displayed in §4.9 are coverage-backed. The real-data effective- F values reported in §4 lie in the range where AR remains close to nominal and uniformly far closer than Wald. The over-identified Angrist–Krueger cell (§4.9) supplies the real-data counterpart to this simulated over-identified failure: there two-stage least squares biases toward ordinary least squares with a spuriously tight standard error, while the AR, Kleibergen-LM, and Moreira-CLR sets are uninformative. We therefore treat the AR set as the primary interval throughout §4 and report Wald only for contrast.

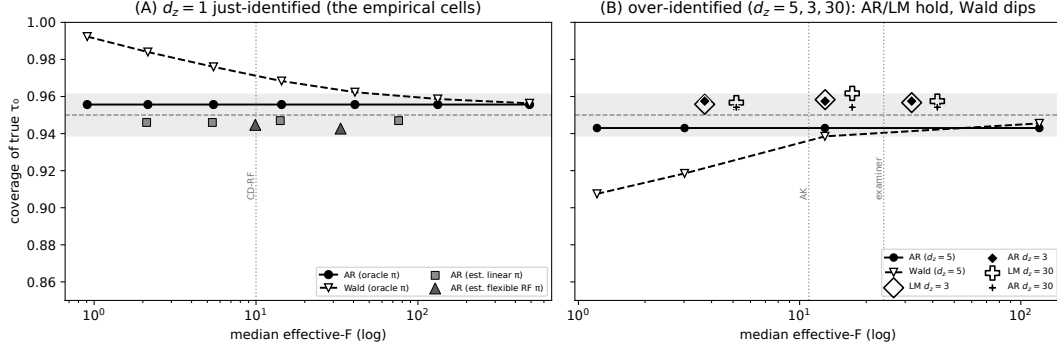


Figure 2: Coverage certificate for the cross-fitted Anderson–Rubin set. Panel (A): just-identified ($d_z = 1$) AR vs. Wald coverage of τ_0 across the concentration sweep under oracle, estimated-linear, and estimated-flexible first stages. Panel (B): over-identified regimes ($d_z = 5, 3, 30$), where AR and Kleibergen–LM hold while Wald dips. Dashed line: nominal 0.95; band: ± 2 Monte-Carlo standard errors.

4.3 Diagnostics and an identification-aware selector

Two further outputs accompany every estimate. First, an **instrument-absorption panel** reports, per learner, R^2_{oof} of each nuisance, the in-sample-minus-out-of-fold overfit gap for $\hat{\pi}$, the residualized-variance ratio $\text{Var}(\tilde{Z})/\text{Var}(Z)$, and a π -calibration slope. These are the quantities that separate the routes of §2.4 and operationalize the interpretation decision tree. Second, although under correctly specified nuisances the prediction-optimal first stage is also concentration-optimal in population (Proposition 3), the two need not coincide at finite n : the relevance g_{ZD} is invariant only in the limit, so among learners that predict Z comparably well out of fold the *realized* residualized concentration still varies. We therefore implement an **identification-aware π -selector** that, among learners statistically tied on out-of-fold predictive loss, chooses the $\hat{\pi}$ with the higher cross-fitted concentration (or effective- F), subject to retaining Çetin’s product-rate condition so that the orthogonal score remains $\sqrt{n}$ -valid. Because the just-identified estimator’s finite-sample bias scales inversely with concentration, this breaks predictive-loss ties toward a less weak-instrument-biased estimate, and it never trades consistency for concentration. A paired known-truth Monte Carlo, run with **correctly specified nuisances**, confirms a modest gain (Figure 3; Table B.3) over 250 paired replications. At a moderate strength (median effective- $F \approx 12$) the rule lowered the mean bias of $\hat{\tau}$ from -0.027 to -0.009 , with the root-mean-square error essentially unchanged ($0.339 \rightarrow 0.334$); in the weakest regime (effective- $F \approx 4$) it roughly halved the root-mean-square error ($2.53 \rightarrow 1.02$) by suppressing catastrophic-tail estimates; and in a low-absorption do-no-harm regime it neither helped nor hurt materially. Anderson–Rubin coverage was essentially identical between the rules (about 0.93). The benefit is genuine: it persists under correctly specified nuisances, so it is not an artifact of the misspecified treatment nuisance the earlier benchmark used. But it is modest: the bias reduction is consistent at moderate and higher strengths and the weak-regime tail control substantial, while the moderate-regime root-mean-square gain is within Monte-Carlo error, and the rule coincides with MSE selection asymptotically (Proposition 3). It is therefore a diagnostically motivated refinement that breaks predictive-loss ties toward concentration at no cost to consistency or coverage, not a replacement for prediction-optimal selection.

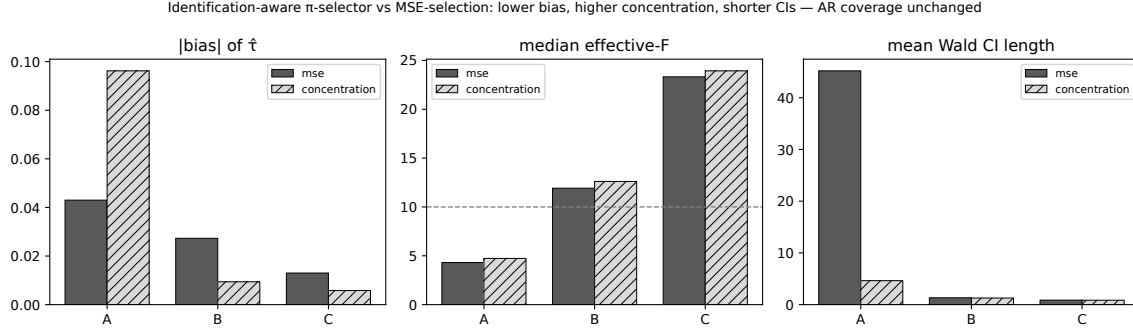


Figure 3: Identification-aware π -selector versus MSE selection across three regimes (250 paired replications, correctly specified nuisances): the concentration tie-break lowers $|\text{bias}|$ of $\hat{\tau}$ at moderate strength, suppresses the weakest-regime catastrophic tails (halving that regime’s RMSE), raises the median effective- F , and shortens the Wald interval, with Anderson–Rubin coverage unchanged.

The selector raises a post-selection-inference question: because $\hat{\pi}$ is chosen to maximize concentration on the same data used for the Anderson–Rubin set, could the selection invalidate the reported coverage? It does not, for two reasons. First, the rule is a *guarded tie-break*: it chooses only among learners within a multiplicative guardrail of the out-of-fold-MSE-best, which are statistically indistinguishable as predictors of Z , so the selected $\hat{\pi}$, and hence the residualized moment, is barely perturbed from the MSE choice. Second, we verify this directly with a matched split-sample experiment: on a held-out inference half we compute the AR coverage of τ_0 both when the first-stage learner is selected on that same half (in-sample) and when it is selected on a disjoint half (which yields an honestly post-selection-valid interval). Across the three regimes the two coincide: the selection distortion (in-sample minus split-sample coverage) is at most 0.008 in absolute value, and exactly zero for the concentration rule at every regime. The concentration tie-break therefore introduces no coverage distortion; the mild departure from nominal that both modes inherit (about 0.92–0.94) is the same finite-sample generated-instrument effect recorded in §4.2, not a selection effect.

4.4 OHIE: a randomized negative control

We now turn from the inference layer to the cells themselves, reading each against the design of §4.1, beginning with the randomized negative control. The Oregon Health Insurance Experiment (Finkelstein et al. 2012) assigned Medicaid eligibility by lottery, so the instrument Z (lottery selection) is randomized conditional on household size. By §2.4, $\pi_0(W)$ is then constant within the conditioning strata and no learner can absorb or dilute Z . We take D to be ever-enrollment in Medicaid, Y a twelve-month survey outcome (primary-care visits), include the required household-size dummies in W , and cluster on household. Following Finkelstein et al. (2012), we take the **weighted-responder sample as the primary specification**: the 23,441 twelve-month survey responders with complete analysis variables (out of the 23,741 responders to that survey), weighted by their survey weights.

The prediction held. In the weighted-responder primary specification the residualized effective- F was approximately 2,100 and stayed flat across the learner library from linear to random forest to gradient boosting (Table 2; Figure 4), and the estimate was stable at $\hat{\tau} \approx 1.04$ additional primary-care visits per complier (ordinary least squares 1.20, two-stage least squares 1.03; the unweighted estimate is a slightly lower 0.97). The absorption diagnostics confirmed the mechanism: every learner, including the random forest, had $R_{\text{of}}^2(\hat{\pi}; Z) \approx 0$ and $\text{Var}(\tilde{Z})/\text{Var}(Z) \approx 1$, so there was nothing to absorb. A collapse here would have indicated a finite-sample DML-IV pathology; its

absence identifies the schooling collapse below as a weak-instrument artifact rather than a property of using machine learning.

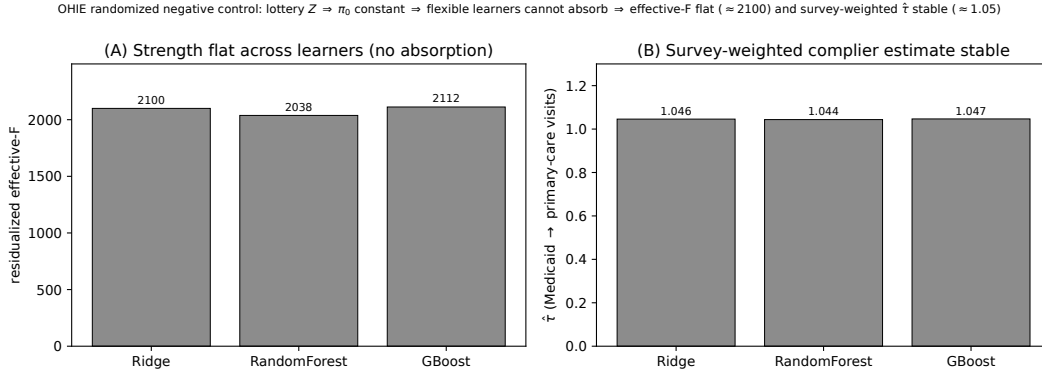


Figure 4: OHIE randomized negative control. Because the lottery makes π_0 constant, no learner can absorb Z : the residualized effective- F stays flat (≈ 2100) across the learner library (A) and the survey-weighted complier estimate $\hat{\tau}$ is stable (≈ 1.04) (B). Values are the frozen Table 2/3 weighted-primary figures.

Because the lottery raises twelve-month survey response (39.6% among the selected versus 26.5% among the non-selected), the survey sample is itself selected on treatment. Lee (2009) trimming bounds, which trim the over-responding lottery arm by the implied 33%, place the complier effect on visits within a wide interval $[-4.0, 4.3]$ that contains zero (Table B.5): the *outcome* effect is not robustly point-identified against worst-case differential non-response. This sensitivity attaches to the survey outcome, not to the cell’s role here. That role, a negative control for first-stage concentration, is unaffected: the effective- F stays in the thousands and flat across learners whatever the outcome’s selection, which is the only property the mechanism argument relies on.

4.5 Examiner leniency: overfit dilution

The examiner cell uses Miami-Dade pretrial data from the Chyn, Frandsen, and Leslie (2025) replication package, with Y a conviction indicator, D pretrial release, and a leave-one-shift-out (cluster-jackknife) judge-lenieny instrument, clustering on the judge-by-shift assignment and absorbing court-by-time fixed effects into W . Because pretrial release is the sign-flipped complement of detention, a negative $\hat{\tau}$ means detention raises conviction. Here the first stage is a genuinely flexible function of judge, court, and time, so overfit dilution of the instrument is an economically real danger; machine-learned instruments in this design class are known to carry finite-sample hazards (Angrist and Frandsen 2022), and the diagnostics below identify which hazard operates here.

We first validated the instrument against the published Stata implementation: our cluster-jackknife first-stage coefficient was 0.319 against their 0.314, and our just-identified cluster-jackknife estimate was about -0.43 against their reported -0.432 , with a covariate-adjusted DML-linear estimate of -0.51 (coupled-nuisance specification, matching their pipeline) reproducing their cluster-jackknife -0.50 ; the decoupled-nuisance first stage reported in Table 2 gives a slightly attenuated DML-ridge $\hat{\tau} \approx -0.45$. The pipeline thus reproduces an independent reference implementation’s estimates.

The flexible first stage then revealed the **dilution** channel rather than absorption. The random-forest first stage could not predict the orthogonalized leniency instrument out of fold ($R_{\text{out}}^2(\hat{\pi}; Z) < 0$) but overfit it in sample, and cross-fitting turned that overfit into noise: $\text{Var}(\tilde{Z})/\text{Var}(Z) = 1.48$, above

one, and the residualized effective- F fell from 25.5 under a ridge first stage to 15.0 under the random forest (Figure 5). The estimate itself moved little (the linear and restricted estimates reproduce the published cluster-jackknife value), and a stability diagnostic showed that the per-fold and pooled estimates diverge under the random forest, flagging the flexible first stage as the unreliable choice for this cell.

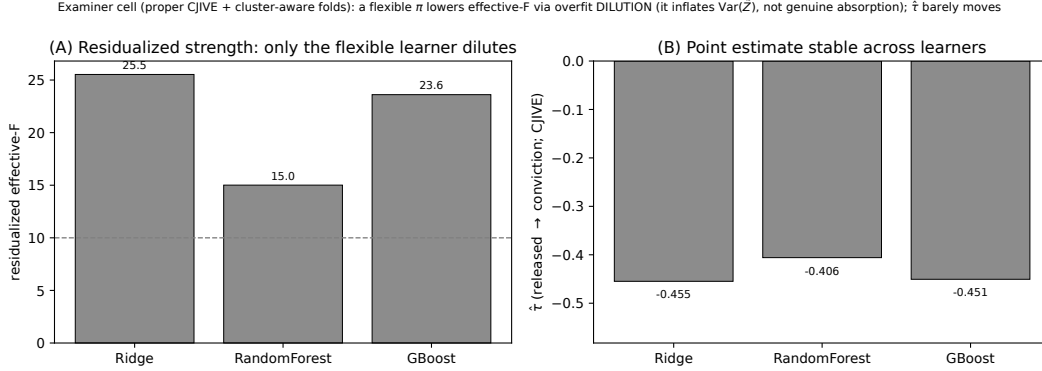


Figure 5: Examiner cell (proper CJIVE, cluster-aware folds). Panel (A): the residualized effective- F drops sharply (25.5 $\rightarrow$ 15.0) under the flexible random-forest first stage via overfit dilution ($\text{Var}(\tilde{Z})/\text{Var}(Z) = 1.48$), not genuine absorption. Panel (B): the point estimate $\hat{\tau}$ is stable across learners. Values are the frozen Table 2/3 figures.

4.6 CollegeDistance: a leakage-corrected cautionary cell

The CollegeDistance cell (Rouse 1995) uses distance to the nearest college as the instrument; it is the application in which Çetin (2026) documents the collapse. It is the one cell where applying our own cluster-aware cross-fitting changes the conclusion, and it shows why that discipline matters. The outcome is a state-level average wage that takes only 41 distinct values and is determined by the state-level controls (region, unemployment, tuition). Under ordinary shuffled cross-validation the same state appears in training and held-out folds, so a flexible first stage “predicts” the instrument by memorizing the state: the random forest then shows out-of-fold $R_{\text{of}}^2(\hat{\pi}; Z) = 0.78$ and the residualized instrument appears strongly absorbed, with the effective- F apparently collapsing. With cluster-aware folds on the state proxy this reverses: R_{of}^2 falls to 0.02 with a large in-sample overfit gap (0.86), the residualized-variance ratio sits just below one ($\text{Var}(\tilde{Z})/\text{Var}(Z) = 0.97$), and the effective- F does not collapse. The apparent genuine absorption was geographic fold leakage; with the leakage removed the cell is simply weak. The cell is therefore diagnosed by the overfit gap rather than by a variance ratio above one, and it is neither a clean absorption nor a clean dilution exemplar.

Two further features make this a cautionary illustration rather than a clean mechanism cell. First, the outcome is region-determined (outcome nuisance $R_{\text{of}}^2 \approx 0.93$), so it is not a clean economic estimand. Second, because each nuisance is fit by the same learner, the linear specification pairs the first stage with a badly misspecified outcome model ($R_{\text{of}}^2 = 0.12$ against the 0.93 achievable), which biases the linear estimate through the *outcome* nuisance rather than the first stage; the linear and random-forest AR sets are then nearly disjoint ($[0.02, 0.10]$ versus $[-0.03, -0.01]$) because the nuisances, not the instrument, differ (once a nuisance is misspecified, the invariance of Proposition 1 no longer applies, and different specifications legitimately produce different estimates). We therefore report CollegeDistance as a leakage-and-misspecification illustration that motivates cluster-aware cross-fitting and separate nuisance learners, and we use the known-truth Monte Carlo (where π_0 is a

designed nonlinear function, τ_0 is known, and the nuisance specification can be controlled) to isolate the treatment-*nuisance* specification effect and to show (Proposition 3) that no genuine-absorption channel operates under correct nuisances (§4.8). Indeed, once the outcome and treatment nuisances are fit by their own MSE-best learners rather than coupled to the first-stage learner, the ridge and random-forest first stages agree: every first stage yields $\hat{\tau} \approx -0.01$ with overlapping Anderson–Rubin sets, confirming that the original disagreement was the coupled (misspecified-linear) outcome nuisance, not the instrument.

4.7 Card: a clean schooling cell, and a within-instrument concentration ladder

The Card (1995) cell replaces CollegeDistance’s state-level outcome with an individual-level one: Y is log hourly wage, D is years of education, and Z indicates growing up near a four-year college (`nearc4`), so the estimand is a genuine return to schooling. With the standard controls (experience and its square, race, region, and urban-residence indicators), on the full sample of 3,010 men, the just-identified two-stage least-squares estimate is $\hat{\tau} = 0.13$ (SE 0.05), the canonical Card magnitude. The cell behaves as Proposition 1 predicts: the decoupled-*nuisance* estimate is **stable across the first-stage library** (0.132 linear, 0.139 random forest, 0.126 gradient boosting), with the residualized effective- F near 15–17 and bounded, overlapping Anderson–Rubin sets (about $[0.03, 0.27]$). Here `nearc4` retains residualized strength, so a flexible first stage neither absorbs nor dilutes it.

Card also affords a controlled view of how the residualized concentration is *built*, because the same instrument can be made strong or weak by the conditioning set. Holding the instrument fixed (`nearc4`, $d_z = 1$) and the sample fixed (the 2,182 men with the full background battery observed), we climb a control-richness ladder: lean controls, then adding family background, parental education, the knowledge-of-the-working-world score, and finally a degree-two expansion. The residualized effective- F falls from about 7 to about 3 (non-monotonically, with an uptick at the +KWW rung; the random-forest ladder is monotone, $6.2 \rightarrow 1.6$; Table B.4), and the Anderson–Rubin set widens from bounded ($[-0.19, 0.26]$) to the whole real line. The decomposition is itself diagnostic: across the ladder the residualized-instrument variance $\text{Var}(\tilde{Z})$ and the first-stage $R_{\text{oof}}^2(\hat{\pi}; Z)$ are essentially flat ($\text{Var}(\tilde{Z}) \approx 0.165\text{--}0.17$, $R_{\text{oof}}^2 \approx 0.20\text{--}0.23$), so the loss is *not* instrument absorption; what falls is the first-stage slope $\text{Cov}(\tilde{Z}, \tilde{D})/\text{Var}(\tilde{Z})$, from 0.26 to 0.14. The background controls absorb the Z – D covariance (the part of `nearc4`’s relevance mediated by family background and region) through the *treatment* nuisance $r(W)$, a route to concentration loss distinct from the π -channels of §2.4. Notably the point estimate does not visibly collapse: $\hat{\tau}$ drifts only within approximately $[0.04, 0.07]$ along the entire linear- π ladder (within $[0.04, 0.10]$ for the random-forest ladder), even as the instrument becomes uninformative. The identification-robust set tells the story the point estimate conceals: once the residualized concentration is gone, the Anderson–Rubin set is the whole line, exactly as weak-instrument theory requires.

This ladder reconciles our clean cell with the returns-to-schooling literature’s mixed experience of Card. In a richer specification the joint first-stage F is 4.69 and the estimate is an uninformative null (two-stage least squares -0.03 , decoupled DML -0.04). That specification uses two proximity instruments (`nearc4` and `nearc2`), the full background battery, and a degree-two expansion, on the restricted 2,182-man sample. This is the specification under which Card is most often judged weak in that literature; our pipeline reproduces its first-stage F (4.69) and two-stage least-squares estimate to the reported digits, the DML estimate (a π -library and residualization choice apart) being a similar uninformative null. The gap between this null and the canonical 0.13 tracks the ladder: restricting to the background-complete subsample roughly halves the concentration (effective- F from about 16 to 7), and the richer controls roughly halve it again (single-instrument effective- F about

3, already below the two-instrument 4.69 and well into the uninformative-null regime). Adding the second proximity instrument `nearc2` raises the joint two-instrument F back to 4.69 (adding an instrument mechanically increases the joint first-stage statistic), but every step past the lean baseline leaves the design too weak to inform τ . Card is therefore not a contradictory data point but the low-concentration end of a single concentration ladder. Which specification to report is itself an identification question. It is answered by the concentration object C and read off the Anderson–Rubin set, not by an a priori verdict on whether the instrument is “good.”

4.8 Cross-cell synthesis

Across the real-data cells, flexible nuisance learning leaves the residualized first stage intact wherever the instrument carries genuine residualized strength; the apparent erosions arise from finite-sample estimation (overfit dilution) or from the conditioning set and the treatment–nuisance specification, never, in population, from a correctly specified flexible first stage. The residualized first stage is flat under randomization (OHIE: effective- F near 2100, $R_{\text{of}}^2 \approx 0$ for every learner) and, importantly, also flat for a real but moderately weak schooling instrument once the cross-fitting is disciplined (Card: effective- F near 15, $\hat{\tau}$ stable at 0.13). The Card cell adds a complementary lesson: holding the instrument and the learner fixed, enriching the controls alone erodes the residualized concentration (here through the first-stage covariance rather than instrument absorption, §4.7), so that concentration is a property of the conditioning set, not of the instrument in the abstract, and the identification-robust set tracks it even where the point estimate does not. The first stage erodes through *overfit dilution* in the examiner cell, where the leniency instrument is itself a high-dimensional estimated object ($R_{\text{of}}^2 < 0$, $\text{Var}(\tilde{Z})/\text{Var}(Z) = 1.48$, effective- F $25.5 \rightarrow 15.0$). The one cell that appeared to show genuine absorption, CollegeDistance, was geographic fold leakage; corrected, it is a weak/ambiguous cell whose flexible-learner movement is an overfitting artifact, diagnosed by the in-sample overfit gap rather than by a variance ratio above one. No real-data cell instantiates *genuine* absorption.

The known-truth Monte Carlo settles what the apparent “absorption” is. With a designed nonlinear π_0 and known τ_0 , a random forest recovers the nonlinear first stage out of fold ($R_{\text{of}}^2(\hat{\pi}; Z) = 0.50$ against a linear 0.18; $\text{Var}(\tilde{Z})/\text{Var}(Z) = 0.50$ against 0.82), and both estimators are essentially unbiased over 100 replications ($|\text{mean bias}| \leq 0.02$ under either nuisance specification, within about one Monte-Carlo standard error of zero). Under the correct r this is Proposition 1’s prediction; under the misspecified r , where Proposition 1’s hypotheses fail, the flexible first stage is consistent and keeps the residualized moment centered (Appendix A.3, Remark). The linear- π arm under the misspecified r is covered by neither result; its near-zero bias here is an empirical feature of this design, whose linear g_0 makes the outcome- and treatment–nuisance misspecifications cancel in the bracket, and it would not be guaranteed under a nonlinear g_0 . The effective- F , however, turns on how the *treatment* nuisance is specified, and this is decisive. Under a correctly specified $r = \mathbb{E}[D | W]$ the flexible first stage attains the **higher** concentration (effective- F about 57 against 34 for a ridge π), exactly as Proposition 3 predicts. The “absorption” ordering, in which the flexible first stage shows the *lower* effective- F (about 97 against 155), appears only when $\hat{\tau}$ is fit by a linear learner that underfits the nonlinear $\mathbb{E}[D | W] = r_0 + f_s \pi_0$ (with f_s the design’s first-stage loading): the unmodeled first-stage structure remains in $\tilde{D}$, and the W -predictable instrument variation that a flexible π removes is correlated with that structure. The drop is therefore a treatment–nuisance specification artifact, not a population cost of flexible π ; it is removed by specifying r flexibly and shrinks as n grows. What the simulation establishes cleanly is thus the *absence* of a genuine-absorption channel and the diagnostic primacy of the treatment–nuisance specification.

The synthesis (Figure 1) is twofold. First, correctly specified flexible learning does not erode the residualized first stage in population (Proposition 3); the realized losses trace to overfit dilution (a

finite-sample estimation pathology, the examiner cell), to the conditioning set (the Card ladder, through $r(W)$), or to an under-specified treatment nuisance (the Monte Carlo); the sign of R_{oof}^2 and the residualized-variance ratio separate them. The losses are absent where the instrument is randomized (OHIE) or retains residualized strength (the Card baseline). Second, what cross-fitting discipline changes is the *diagnosis*, not the point estimate. With cluster-aware folds and decoupled nuisances, the one apparent genuine-absorption collapse (CollegeDistance) is revealed as geographic fold leakage (its out-of-fold R_{oof}^2 falls from 0.78 to 0.02 and its effective- F is corrected), while the flexible-first-stage point estimate itself stays near zero; the discipline relocates the cause rather than rescuing the estimate. The practical lesson is that a near-zero estimate under flexible learning is a statement about first-stage concentration and inference, not a defect of machine learning per se. Such an estimate is to be read with the absorption diagnostics, an identification-robust interval, and disciplined cross-fitting.

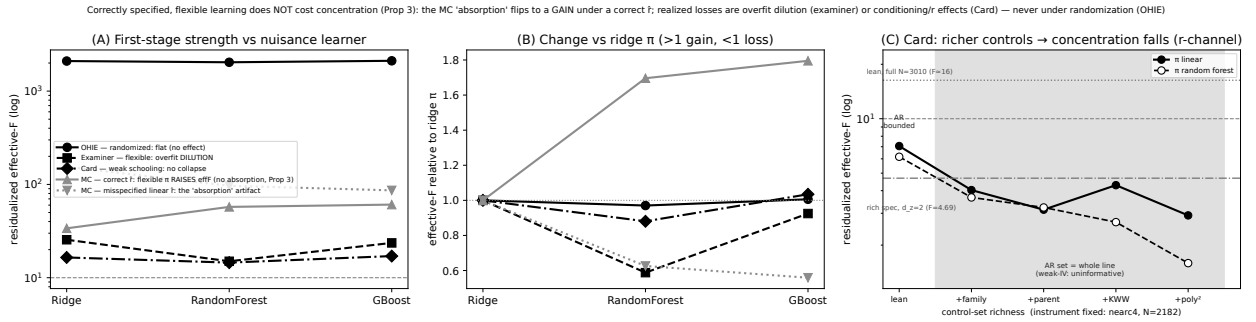


Figure 1: Cross-cell synthesis. (A) residualized effective- F by nuisance learner across cells and the Monte-Carlo two-specification contrast; (B) the same relative to the ridge- π baseline (> 1 gain, < 1 loss); (C) the Card control-richness ladder (instrument fixed) along which richer controls lower concentration through the treatment channel $r(W)$. Correctly specified flexible learning never costs concentration (Proposition 3); realized losses are overfit dilution (examiner), conditioning/ r effects (Card), or an under-specified treatment nuisance (Monte Carlo).

4.9 Angrist–Krueger: over-identified weak instruments

The four cells above are just-identified ($d_z = 1$), where the Anderson–Rubin, Kleibergen-LM, and Moreira-CLR sets coincide (§3.2). The Angrist and Krueger (1991) quarter-of-birth design supplies the over-identified counterpart and the real-data instance of the over-identified weak-instrument failure that §4.2 documents in simulation. On the full $N = 329,509$ men in the Census extract, Y is log weekly wage, D is years of schooling, the controls W are year-of-birth and state-of-birth dummies, and standard errors are clustered on state of birth; because W is a saturated set of dummies, the linear nuisance is exactly $\mathbb{E}[\cdot | W]$ and flexible learners add nothing, so the cell isolates the over-identified *inference*, not the absorption mechanism. We run two instrument sets: the three quarter-of-birth dummies ($d_z = 3$) and their interactions with year of birth ($d_z = 30$), the Bound, Jaeger, and Baker (1995) many-weak specification.

With the three quarter-of-birth dummies the cell behaves as a clean over-identified design. Two-stage least squares and the cross-fitted DML estimate agree at $\hat{\tau} = 0.108$ (Wald SE 0.017 and 0.020), the canonical Angrist–Krueger return to schooling and well above the ordinary-least-squares 0.067; the effective- F is 11.4; and the three identification-robust sets (now genuinely distinct objects at $d_z > 1$) are bounded and mutually close (Table B.6): AR [0.051, 0.148], Kleibergen-LM [0.057, 0.139], Moreira-CLR [0.058, 0.138]. The Hansen–Sargan over-identification test does not reject ($J = 2.84$, 2 d.f., $p = 0.24$), so the three instruments are mutually consistent. This is the first

cell to exercise the Kleibergen-LM, Moreira-CLR, and Hansen-Sargan procedures that §3.2 reserves for the over-identified case, and they behave as the theory requires.

Saturating the instrument to the thirty quarter-of-birth $\times$ year-of-birth interactions turns the design into the many-weak case, and the two faces of that problem appear together. Two-stage least squares biases toward ordinary least squares ($\hat{\tau} = 0.087$, between the 0.108 of the clean design and the 0.067 of OLS), with a *tighter* Wald standard error (0.011) that reads as spurious precision, exactly the Bound-Jaeger-Baker pathology. The identification-robust sets, by contrast, report the design honestly: the AR set is the whole line and the Kleibergen-LM and Moreira-CLR sets are unbounded, so none of the three localizes τ . The Hansen-Sargan test still does not reject ($J = 25.2$, 29 d.f., $p = 0.67$): the instruments are valid, merely weak, and the over-identification test has little power to say more. Notably the per-instrument effective- F (12.3) sits above the rule of ten even as the thirty-dimensional design cannot inform τ . This is the over-identified analogue of the §4.2 lesson that the effective- F screens strength while the identification-robust set, not an F -threshold, certifies inference. The over-identified weak-instrument failure that §4.2 establishes in simulation (Wald misleads, the robust set does not) is thus reproduced on the most heavily studied weak-instrument dataset in econometrics. The inference is verified rather than assumed: a coverage simulation calibrated to these two operating points (§4.2) confirms that at both $d_z = 3$ and $d_z = 30$ the cross-fitted AR, Kleibergen-LM, and Moreira-CLR sets hold near-nominal coverage (0.953–0.962) and the Hansen-Sargan J test has nominal size (0.044).

5. Discussion

Our results recast a documented fragility of DML-IV as a problem of inference and design rather than of identification. When a flexible first stage drives a PLIV estimate toward zero with a small standard error, the evidence here suggests that the right reading is not a precise zero but a loss of residualized first-stage concentration, which invalidates the Wald interval while leaving the estimand unchanged. The practical implication is concrete: in any DML-IV application the analyst should report the residualized effective- F and a cross-fitted Anderson-Rubin set by default, and should read the instrument-absorption diagnostics before interpreting a near-zero estimate, because the diagnostics, not the point estimate, reveal whether the first stage has weakened.

The diagnostic distinction is more than a taxonomy, because the sources call for different responses. Correctly specified, flexible learning carries no concentration cost in population (Proposition 3): the prediction-optimal first stage is also concentration-optimal, so a flexible $\hat{\pi}$ recovering a nonlinear π_0 is to be preferred, not feared. A realized loss is then one of two things. The first is overfit dilution: the first stage cannot predict the instrument out of fold yet injects cross-fitted noise, an estimator pathology that a restricted or more heavily regularized first stage removes at no cost to consistency. The second is a treatment-nuisance and conditioning effect: a richer conditioning set genuinely lowers conditional relevance (a modeling choice), and an under-specified $\hat{\tau}$ can manufacture an apparent absorption that a flexibly specified $\hat{\tau}$ removes. The sign of the out-of-fold predictive fit, the residualized-variance ratio, and the behavior under a flexibly specified treatment nuisance indicate which case obtains, and therefore whether the response is to regularize the first stage, to respecify the treatment nuisance, or to accept a genuinely weaker conditional design.

These findings sit between several literatures. The documented sensitivity, the proposed absorption reading, and the concentration quantity are Çetin's; our contribution is to execute the weak-instrument-robust inference he flags and to supply a design that varies first-stage strength so the mechanism can be identified rather than assumed. The inference layer draws on the weak-instrument tradition (the effective- F screen of Montiel Olea and Pflueger and the identification-robust sets of

Anderson and Rubin and their refinements), applied here to a cross-fitted, generated instrument rather than a fixed one. The examiner cell connects to the examiner-design literature and its cluster-jackknife estimator, against which we validate our implementation. Proposition 3 is related to, but distinct from, the classical optimal-instruments theory (Amemiya 1974; Chamberlain 1987; Newey 1990). That theory chooses the instrument *transformation* $h(Z, W)$ to attain the efficiency bound under strong identification; here the instrument is held fixed and the choice object is the *residualization* π . Within the residualization class the two aims coincide under conditional homoskedasticity of the score, where $\text{Var}(\hat{\tau}) \propto 1/(nC)$ is exact (Appendix A.1); under heteroskedasticity C ranks weak-identification strength rather than the sandwich variance, which is the sense in which Proposition 3 is a weak-identification result rather than an efficiency result. We develop this delineation formally in follow-up work. What the cross-cell contrast adds is evidence that flexible nuisance learning is not uniformly hazardous: under randomization the population absorption channel is closed (Proposition 3) and no realized loss appears in the OHIE cell, and where flexible learning does harm, standard diagnostics localize the cause.

The conclusions are robust along the dimensions a referee would check, and the checks are reported within the cells rather than relegated to a separate battery. Across the nuisance-learner library the estimate is stable wherever the residualized instrument retains strength: OHIE holds effective- $F \approx 2100$ and $\hat{\tau} \approx 1.04$ flat across ridge, random-forest, and gradient-boosting first stages, and Card holds $\hat{\tau} \approx 0.13$ across the library (Table 2, §4.7). Where the instrument does not retain strength, in the examiner random-forest first stage, the divergence of the per-fold (DML1) and pooled (DML2) estimates and the across-split variability are themselves the overfit-dilution signature of §4.5, flagged rather than hidden. Control-set and instrument-set sensitivity is mapped explicitly and read off the identification-robust interval: the Card control-richness ladder traces the concentration from a bounded AR set to the whole line as the conditioning set grows richer (§4.7), and the Angrist–Krueger cell contrasts three versus thirty instruments (§4.9). The cross-fitting protocol is stress-tested rather than assumed: cluster-aware folds versus plain K -fold change the CollegeDistance diagnosis (the geographic-leakage correction, §4.6), and decoupling the outcome and treatment nuisances from the first-stage learner removes the coupled-learner artifact. Finally, the cross-fitted Anderson–Rubin coverage holds not only on the linear simulation design but under nonlinear outcome and treatment structural functions estimated by random forests (§4.2), so its validity does not rest on linear nuisances.

Several limitations bound these conclusions. While the concentration result is a theorem (Proposition 3), the *attribution* of a realized loss to overfit dilution versus treatment-nuisance misspecification is an operational characterization supported by diagnostics, and the finite-sample boundary between a correctly recovered nonlinear first stage and overfitting is not sharp. The estimands are complier local average treatment effects with binary treatments, so the partially linear specification is an approximation. The coverage certificate is calibrated to the regimes the cells occupy (effective- F from about one to about five hundred in the simulation sweep, instrument dimension up to thirty, the over-identified Angrist–Krueger regimes included) and does not claim universal validity; in particular, a second-order product of nuisance estimation errors enters the cross-fitted Anderson–Rubin moment with a factor of the tested τ , so the far endpoints of the set lean hardest on the product-rate condition, and a locally robust corrected moment that suppresses this term at every tested value is left to future work. Finally, the CollegeDistance cell carries a region-determined outcome and therefore serves as a leakage-and-misspecification stress test rather than a clean estimand; the clean schooling estimand is supplied by the Card cell (§4.7).

These limitations point to concrete next steps. A locally robust corrected Anderson–Rubin moment, in the sense of Chernozhukov, Escanciano, Ichimura, Newey, and Robins (2022), would make the identification-robust set uniformly valid rather than verified only on calibrated regimes.

A formal dominance result for the identification-aware selector, beyond the Monte Carlo evidence reported here, would complete the constructive part of the analysis.

6. Conclusion

We have argued that the collapse of a DML-IV estimate under flexible nuisance learning is a loss of residualized first-stage concentration, not of identification. With a correctly specified treatment nuisance it is not an inherent cost of flexible learning at all: the prediction-optimal first stage then maximizes the population concentration (Proposition 3). The realized losses run through diagnosable sources that standard diagnostics tell apart: overfit dilution, the conditioning set, and treatment- nuisance specification. We executed the weak-instrument-robust inference this requires and verified, in a known-truth Monte Carlo, that a cross-fitted Anderson–Rubin set keeps coverage close to nominal and far closer than Wald across the regimes our applications occupy, where Wald either under-covers (over-identified) or masks a severely biased estimate behind an inflated standard error (just-identified). A strength-as-treatment design, from a randomized lottery through a high-dimensional examiner instrument to a weak proximity instrument, located overfit dilution in the examiner data and the conditioning-set route in the Card ladder, and confirmed in the randomized control that neither operates, with the simulation showing the apparent absorption to be a treatment- nuisance artifact; and an identification-aware first-stage selector modestly reduced the estimator’s finite-sample bias at moderate strength and its tail risk in the weakest regime by breaking predictive-loss ties toward a more concentrated first stage.

The practical takeaway is narrow and actionable. In the partially linear IV model, a near-zero estimate with a small standard error under flexible nuisance learning should be read, absent a strong residualized first stage, as a weak-instrument signal rather than a precise zero, and reported with an identification-robust interval and the absorption diagnostics that locate the source of the collapse. The concentration result is a theorem (Proposition 3), while the attribution of a realized loss to its source is diagnostic; the simulation-verified coverage is established on the regimes the data occupy rather than universally; and clean external corroboration is supplied by the over-identified Angrist–Krueger cell (§4.9), where two-stage least squares biases toward ordinary least squares while the identification-robust sets correctly report the many-weak design as uninformative. Within those bounds, the analysis turns a documented fragility of machine-learning-based IV estimation into a measurable, reportable, and partly correctable property of the first stage.

Appendix A. Propositions 1–3: sources and proofs

Notation is as in §2.1: the PLIV model is $Y = \tau_0 D + g_0(W) + \zeta$ with $\mathbb{E}[\zeta \mid Z, W] = 0$; the nuisances are $\mu_0 = \mathbb{E}[Y \mid W]$, $r_0 = \mathbb{E}[D \mid W]$, $\pi_0 = \mathbb{E}[Z \mid W]$; residuals at the truth are $\tilde{Z}_0 = Z - \pi_0$, $\tilde{D}_0 = D - r_0$, and at a general first stage $\tilde{Z} = Z - \pi$. The just-identified estimator solves the empirical analogue of $\mathbb{E}[(Z - \pi)\{(Y - \mu_0) - \tau(D - r_0)\}] = 0$; for $d_z > 1$ the optimally weighted GMM root is $\hat{\tau} = (g'_{ZD}\Omega^{-1}g_{ZY})/(g'_{ZD}\Omega^{-1}g_{ZD})$ with $g_{ZD} = \mathbb{E}[\tilde{Z}\tilde{D}]$, $g_{ZY} = \mathbb{E}[\tilde{Z}\tilde{Y}]$, $\Omega = \mathbb{E}[\tilde{Z}\tilde{Z}']$. Propositions 1 and 2 formalize results whose ingredients are available in the cited sources, and A.1–A.2 accordingly record statements, sources, and proof sketches only; Proposition 3 is, to our knowledge, new here, and A.3 gives its complete proof.

A.1 Proposition 1: sources and proof sketch

Proposition 1 formalizes the estimand invariance implicit in the consistency theory for orthogonalized PLIV moments; the ingredients are those of Chernozhukov et al. (2018) and Çetin (2026), so we record only the skeleton. We record it at all because the π -invariance of the slope is used repeatedly in the text. At $\tau = \tau_0$ the bracket of the moment collapses to the structural error: $\mu_0 - \tau_0 r_0 = \mathbb{E}[Y - \tau_0 D \mid W] = g_0$ by the tower property, so $m(\tau_0; \pi_*) = \mathbb{E}[(Z - \pi_*)\zeta]$. Instrument exogeneity kills both pieces of the population moment $m(\tau_0; \pi_*)$: $\mathbb{E}[Z\zeta] = \mathbb{E}[Z\mathbb{E}[\zeta \mid Z, W]] = 0$ (the **full** conditioning on Z is needed here, since Z is not W -measurable), and $\mathbb{E}[\pi_*(W)\zeta] = 0$ for any W -measurable π_* . The moment is affine in τ with slope $-g$, and $g = \mathbb{E}[(Z - \pi_*)\tilde{D}_0] = \mathbb{E}[\tilde{Z}_0\tilde{D}_0]$ is invariant to π_* because $\mathbb{E}[\tilde{D}_0 \mid W] = 0$; the root is therefore unique and equal to τ_0 for **every** admissible first stage. Cross-fitting transports the population root to the estimator under L^2 -consistent nuisances by the standard argument (Chernozhukov et al. 2018; Çetin 2026), and for $d_z > 1$ the identity $g_{ZY} = \tau_0 g$ makes the optimally weighted GMM root equal to τ_0 for any positive-definite weight. Consistency uses only L^2 nuisance convergence; the $\sqrt{n}$ -asymptotic normality invoked in §2.1 and §3 additionally requires the product-rate condition of Çetin (2026), which is not used for the present claim. The variance comparison $\text{Var}(\hat{\tau}) \propto 1/(nC)$ across first stages is likewise a strong-identification statement established at a consistent first stage; at a limit $\pi_* \neq \pi_0$ the score loses its orthogonality in the (μ, r) directions, and we do not develop distribution theory there. The proportionality is exact under conditional homoskedasticity of the score, where $\Sigma(\pi) \propto \Omega(\pi)$; under heteroskedasticity C ranks weak-identification strength rather than the sandwich variance itself.

A.2 Proposition 2: sources and proof sketch

Part (i) is immediate: under $\tilde{Z} \mapsto c\tilde{Z}$ the numerator moment scales as $c^2\Sigma$ and the relevance as $c g_{ZD}$, so $V = \Sigma/g_{ZD}^2$ is unchanged. A small standard error thus encodes a small g_{ZD} (limited residualized instrument variation) rather than low noise; this is the reading Çetin (2026) gives of his Table 2. Part (ii) is the classical weak-instrument asymptotics of Staiger and Stock (1997), applied to the cross-fitted residualized moment (Andrews, Stock, and Sun 2019 survey the theory); we state the limits and do not reproduce the derivations. With $u = \tilde{Y} - \tau_0 \tilde{D}$, $\Sigma = \mathbb{E}[\tilde{Z}^2 u^2]$, $\omega^2 = \text{Var}(\tilde{Z}\tilde{D})$, and the local-to-zero sequence $g_{ZD,n} = \gamma/\sqrt{n}$, a joint central limit theorem for the numerator and the centered denominator yields the following limits, all evaluated at the true τ_0 :

$$\hat{\tau} - \tau_0 \Rightarrow \frac{\sqrt{\Sigma} \xi_1}{\gamma + \omega \xi_2}, \quad (\text{A.1})$$

$$\widehat{\text{se}} \Rightarrow \frac{\sqrt{\Sigma}}{|\gamma + \omega \xi_2|} = O_p(1), \quad (\text{A.2})$$

$$\frac{\hat{\tau} - \tau_0}{\widehat{\text{se}}} \Rightarrow \xi_1 \cdot \text{sign}(\gamma + \omega \xi_2), \quad (\text{A.3})$$

with (ξ_1, ξ_2) standard bivariate normal of correlation $\rho = \text{Cov}(\tilde{Z}u, \tilde{Z}\tilde{D})/(\sqrt{\Sigma}\omega)$. The three displays carry the proposition's content: $\hat{\tau}$ is inconsistent and $O_p(1)$, with no $\sqrt{n}$ rate; the standard error is finite and bounded away from zero rather than vanishing; and the t -ratio is $N(0, 1)$ iff $\rho = 0$ and non-pivotal otherwise, so no fixed critical value attains nominal coverage. The failure is the *shape* of (A.3), not the magnitude of (A.2). Two ingredients here are non-classical: $\tilde{Z}$ is a cross-fitted, generated instrument, and the feasible standard error evaluates the residuals at the inconsistent $\hat{\tau}$ rather than at τ_0 , which perturbs Σ by an $O_p(1)$ amount without restoring pivotality. Both are why §4.2 verifies coverage by simulation rather than assuming the classical limits transfer.

Proposition 2(ii) concerns the local-to-zero *sequence*: at any fixed first-stage strength the t -ratio is asymptotically $N(0, 1)$, and the finite-sample coverage certificate of §4.2 measures this sequence's shadow at small effective- F .

A.3 Proof of Proposition 3

This is the paper's central theoretical claim, so we give it in full. **Setup.** The treatment nuisance is correctly specified, $r = r_0$, so $\tilde{D} = D - r_0$ is conditional-mean-zero: $\mathbb{E}[\tilde{D} | W] = 0$, the single property the entire argument rests on. For a square-integrable, W -measurable first stage π (with $\pi_0 = \mathbb{E}[Z | W]$ the truth) write $\tilde{Z} = Z - \pi$, $\tilde{Z}_0 = Z - \pi_0$, and define the residualized concentration

$$C(\pi) = g_{ZD}(\pi)' \Omega(\pi)^{-1} g_{ZD}(\pi), \quad g_{ZD}(\pi) = \mathbb{E}[\tilde{Z}\tilde{D}] \in \mathbb{R}^{d_z}, \quad \Omega(\pi) = \mathbb{E}[\tilde{Z}\tilde{Z}'] \in \mathbb{R}^{d_z \times d_z}.$$

Strategy. $C(\pi)$ is a ratio: a numerator built from the relevance vector $g_{ZD}(\pi)$ and a denominator built from the instrument's second-moment matrix $\Omega(\pi)$. We show the numerator is *constant* in π (Lemma A) while the denominator is *Loewner-minimized* exactly at π_0 (Lemma B); combining the two through monotonicity of the matrix inverse forces $C(\pi) \leq C(\pi_0)$. For how this result relates to, and differs from, the classical optimal-instruments theory (Amemiya 1974; Chamberlain 1987; Newey 1990), see the delineation in §5.

Lemma A (the relevance vector does not depend on π). For every admissible π ,

$$g_{ZD}(\pi) = \mathbb{E}[(Z - \pi)\tilde{D}] = \mathbb{E}[Z\tilde{D}] - \mathbb{E}[\pi(W)\tilde{D}] = \mathbb{E}[Z\tilde{D}] - \mathbb{E}[\pi(W) \underbrace{\mathbb{E}[\tilde{D} | W]}_{=0}] = \mathbb{E}[Z\tilde{D}] =: g,$$

where the third equality is the tower property applied to the W -measurable factor $\pi(W)$, and the fourth uses $\mathbb{E}[\tilde{D} | W] = 0$. Thus $g_{ZD}(\pi) \equiv g$ for all π ; in particular, taking $\pi = \pi_0$, $g = \mathbb{E}[\tilde{Z}_0\tilde{D}]$. *Intuition.* Correct treatment residualization strips every W -predictable feature out of $\tilde{D}$, so whatever W -function $\pi(W)$ one subtracts from Z is uncorrelated with $\tilde{D}$: the identifying covariance lives entirely in the part of Z that *no* first stage can touch. Changing π moves only the noise in $\tilde{Z}$, never its signal.

Lemma B (the residualized second moment is Loewner-minimized at π_0 ; a standard best-prediction fact, recorded for completeness). Decompose the residualized instrument into the true residual plus a deterministic, W -measurable shift,

$$\tilde{Z} = Z - \pi = (Z - \pi_0) + (\pi_0 - \pi) = \tilde{Z}_0 + \Delta, \quad \Delta := \pi_0 - \pi \text{ (} W\text{-measurable)}.$$

Because $\pi_0 = \mathbb{E}[Z | W]$ is the L^2 -projection of Z onto W , its residual is orthogonal to every W -measurable function; concretely $\mathbb{E}[\tilde{Z}_0 | W] = \mathbb{E}[Z - \pi_0 | W] = 0$, so the cross term vanishes:

$$\mathbb{E}[\tilde{Z}_0\Delta'] = \mathbb{E}[\mathbb{E}[\tilde{Z}_0 | W]\Delta'] = \mathbb{E}[0 \cdot \Delta'] = 0.$$

Expanding $\Omega(\pi) = \mathbb{E}[(\tilde{Z}_0 + \Delta)(\tilde{Z}_0 + \Delta)']$ and dropping the two (transpose-equal) cross terms gives the exact Pythagorean decomposition

$$\Omega(\pi) = \mathbb{E}[\tilde{Z}_0\tilde{Z}_0'] + \mathbb{E}[\Delta\Delta'] = \Omega(\pi_0) + \mathbb{E}[(\pi_0 - \pi)(\pi_0 - \pi)'] \succeq \Omega(\pi_0), \quad (\text{A.4})$$

since $\mathbb{E}[\Delta\Delta']$ is a second-moment matrix, hence positive semidefinite, with $\Omega(\pi) = \Omega(\pi_0)$ iff $\mathbb{E}[\Delta\Delta'] = 0$, i.e., $\Delta = 0$ a.s., i.e., $\pi = \pi_0$ a.s. *Intuition.* Residualizing with the true π_0 leaves the minimum-variance residual $\tilde{Z}_0$; replacing π_0 by any other π injects a deterministic W -measurable error Δ that, being orthogonal to $\tilde{Z}_0$, can only *add* spread to the residualized instrument, never remove it.

Step (monotonicity of the matrix inverse). If $A \succeq B \succ 0$ then $A^{-1} \preceq B^{-1}$. Indeed $A \succeq B \succ 0$ is equivalent to $B^{-1/2}AB^{-1/2} \succeq I$, so all eigenvalues of $B^{-1/2}AB^{-1/2}$ are ≥ 1 ; the eigenvalues of its inverse $B^{1/2}A^{-1}B^{1/2}$ are then ≤ 1 , i.e., $B^{1/2}A^{-1}B^{1/2} \preceq I$, which is $A^{-1} \preceq B^{-1}$.

Conclusion. Take first the just-identified case $d_z = 1$, where everything is scalar: g is a number, $\Omega(\pi) = \mathbb{E}[\tilde{Z}^2] \geq \mathbb{E}[\tilde{Z}_0^2] = \Omega(\pi_0)$ by (A.4), and

$$C(\pi) = \frac{g^2}{\Omega(\pi)} \leq \frac{g^2}{\Omega(\pi_0)} = C(\pi_0),$$

because a fixed positive numerator over a larger denominator is smaller; this scalar case is the clearest form of the result. For general d_z , the relevance g is fixed (Lemma A) and $\Omega(\pi) \succeq \Omega(\pi_0) \succ 0$ (Lemma B), so by inverse monotonicity $\Omega(\pi)^{-1} \preceq \Omega(\pi_0)^{-1}$, and therefore

$$C(\pi) = g' \Omega(\pi)^{-1} g \leq g' \Omega(\pi_0)^{-1} g = C(\pi_0), \quad (\text{A.5})$$

the inequality being the statement that $g'(\Omega(\pi_0)^{-1} - \Omega(\pi)^{-1})g \geq 0$. The prediction-optimal first stage π_0 thus **maximizes** the residualized concentration over all square-integrable W -measurable π and every d_z , using nothing but $\mathbb{E}[\tilde{D} | W] = 0$ and orthogonal projection. A consistent flexible learner that recovers a nonlinear π_0 therefore *attains* the maximum concentration: it cannot, in population, lower it.

Equality. When is the bound (A.5) tight? Write $A = \Omega(\pi_0) \succ 0$ and $M = \mathbb{E}[\Delta\Delta'] = \Omega(\pi) - A \succeq 0$, and factor $M = LL'$ (e.g., $L = M^{1/2}$). The Woodbury identity gives the difference of inverses in manifestly positive-semidefinite form,

$$A^{-1} - (A + M)^{-1} = A^{-1}L(I + L'A^{-1}L)^{-1}L'A^{-1},$$

so the concentration gap is a squared length,

$$C(\pi_0) - C(\pi) = g'(A^{-1} - (A + M)^{-1})g = \|(I + L'A^{-1}L)^{-1/2}L'A^{-1}g\|^2 \geq 0,$$

vanishing iff $L'A^{-1}g = 0$, equivalently $MA^{-1}g = 0$, i.e.

$$M\Omega(\pi_0)^{-1}g = 0. \quad (\text{A.6})$$

For $d_z = 1$, $M = \mathbb{E}[\Delta^2] \geq 0$ is a scalar and $\Omega(\pi_0)^{-1}g \neq 0$ (relevance), so (A.6) holds iff $M = 0$, i.e., **iff** $\pi = \pi_0$ **a.s.:** any nonzero perturbation strictly lowers concentration. For $d_z > 1$ the condition is genuinely weaker: $\Omega(\pi_0)^{-1}g$ is the population (optimal-GMM) slope direction, and (A.6) says the added-variance matrix M *annihilates* that direction. A misspecified $\pi \neq \pi_0$ can then be **concentration-neutral**, $C(\pi) = C(\pi_0)$, precisely when the extra residual variance it injects lies entirely in directions orthogonal to the slope (variation that carries no identifying information). In every case, recovering π_0 attains the maximum. ■

Remark (apparent “absorption” requires a misspecified r). The entire argument used $\mathbb{E}[\tilde{D} | W] = 0$, i.e., $r = r_0$. If instead $\hat{r} \not\rightarrow r_0$ (for instance a linear $\hat{r}$ when $\mathbb{E}[D | W] = r_0 + f_s\pi_0$ is nonlinear), then $\mathbb{E}[\tilde{D} | W] \neq 0$, **Lemma A fails**, and $g_{ZD}(\pi)$ becomes π -dependent: a flexible π that strips the W -predictable part of Z now also removes variation correlated with the unmodeled first-stage structure left in $\tilde{D}$, **lowering** C . Yet $\hat{r}$ remains consistent when the first stage is itself consistent: with $\hat{\pi} \rightarrow \pi_0$ the residualized instrument is conditional-mean-zero, so $\mathbb{E}[(Z - \pi_0)(r^* - r_0)] = 0$ for the limit r^* of $\hat{r}$ (and likewise for any outcome-nuisance limit), and the moment still has its root at τ_0 (Proposition 1’s blanket π -invariance, by contrast, requires $(\mu, r) \rightarrow (\mu_0, r_0)$). This

nuisance-misspecification interaction, *not* a property of flexible π , is exactly the regime isolated by the §4.8 Monte Carlo. With estimated but correctly specified nuisances, the realized C can fall below $C(\pi_0)$ only through finite-sample estimation noise (the overfit-dilution channel, signed by $R_{\text{of}}^2(\hat{\pi}; Z) \leq 0$ together with $\text{Var}(\tilde{Z})/\text{Var}(Z) > 1$), which vanishes as $n \rightarrow \infty$. There is no population “genuine absorption” channel under correct nuisances; Proposition 3 closes it off.

Appendix B. Full result tables

These tables report the complete per-cell estimator panels and the Monte-Carlo certificates summarized in the main text; every value is the committed artifact produced by the replication pipeline.

Table B.1: Full estimator panel by cell. For each empirical cell we report OLS, 2SLS, and DML with CV-tuned ridge, random-forest, and gradient-boosted nuisances. The AR column is the 95% Anderson–Rubin / weak-IV-robust set (em-dash where the estimator does not report one).

Cell / Estimator	$\hat{\tau}$	SE	effective- F	AR 95% set	N
<i>OHIE (Oregon Health Insurance Experiment)</i>					
OLS	1.196	0.063	—	—	23,441
2SLS	1.034	0.173	2089.34	—	23,441
DML (ridge)	1.046	0.173	2099.86	[0.705, 1.385]	23,441
DML (RF)	1.044	0.175	2037.73	[0.700, 1.385]	23,441
DML (GB)	1.047	0.172	2112.27	[0.710, 1.385]	23,441
<i>Examiner design (judge leniency)</i>					
OLS	-0.158	0.005	—	—	91,282
2SLS	-0.497	0.189	24.45	—	91,282
DML (ridge)	-0.455	0.142	25.53	[-0.914, -0.060]	91,282
DML (RF)	-0.406	0.174	15.01	[-1.044, 0.106]	91,282
DML (GB)	-0.451	0.146	23.61	[-0.930, -0.046]	91,282
<i>CollegeDistance (Rouse 1995)</i>					
OLS	-0.001	0.001	—	—	4,739
2SLS	0.052	0.020	27.59	—	4,739
DML (ridge)	-0.005	0.004	10.99	[-0.013, 0.002]	4,739
DML (RF)	-0.011	0.005	8.85	[-0.024, -0.004]	4,739
DML (GB)	-0.014	0.005	9.13	[-0.027, -0.006]	4,739
<i>Card (1995) returns to schooling</i>					
OLS	0.075	0.004	—	—	3,010
2SLS	0.132	0.054	13.49	—	3,010
DML (ridge)	0.132	0.053	16.49	[0.031, 0.270]	3,010
DML (RF)	0.139	0.055	14.52	[0.035, 0.291]	3,010
DML (GB)	0.126	0.052	17.06	[0.026, 0.259]	3,010

Table B.2: Coverage certificate: cross-fitted Anderson–Rubin vs. Wald coverage across simulation regimes. Each row is one Monte-Carlo operating point; the AR set retains nominal coverage as the median effective- F falls, whereas Wald coverage degrades. MCSE and Wald coverage are reported where the regime records them (em-dash otherwise). In the clustered-DGP rows the AR column is the *cluster*-robust set (what the examiner/OHIE cells use); the naive heteroskedasticity-only AR under-covers there (0.71 and 0.51), so clustering must be, and is, accounted for. In the survey-weighted rows the AR is the *weighted* set covering the survey-population estimand τ_w ; a naive unweighted AR covers τ_w only 0.56 and 0.07 of the time (it targets the wrong, unweighted estimand τ_u).

Regime	d_z	median eff.- F	AR cov.	AR MCSE	Wald cov.	reps
<i>$d_z = 1$ oracle nuisances</i>						
	1	0.91	0.956	—	0.992	3,000
	1	2.15	0.956	—	0.984	3,000
	1	5.46	0.956	—	0.976	3,000
	1	14.39	0.956	—	0.968	3,000
	1	40.82	0.956	—	0.962	3,000
	1	132.63	0.956	—	0.959	3,000
	1	489.18	0.956	—	0.956	3,000
<i>$d_z = 1$ estimated (linear) nuisances</i>						
	1	2.11	0.946	—	0.979	1,000
	1	5.37	0.946	—	0.971	1,000
	1	14.11	0.947	—	0.963	1,000
	1	75.98	0.947	—	0.954	1,000
<i>$d_z = 1$ estimated (RF, nonlinear) nuisances</i>						
	1	9.90	0.945	0.0059	0.965	1,500
	1	33.30	0.943	0.0060	0.957	1,500
<i>$d_z = 1$ nonlinear nuisances</i>						
	1	2.85	0.976	0.0097	0.988	250
	1	9.81	0.968	0.0111	0.972	250
<i>$d_z = 5$ over-identified</i>						
	5	1.21	0.943	—	0.907	2,000
	5	3.02	0.943	—	0.918	2,000
	5	13.02	0.943	—	0.939	2,000
	5	121.25	0.943	—	0.946	2,000
<i>$d_z = 3$ over-identified</i>						
	3	4.23	0.958	0.0058	—	1,200
	3	14.82	0.958	0.0058	—	1,200
	3	36.49	0.958	0.0058	—	1,200
<i>$d_z = 30$ over-identified</i>						
	30	4.58	0.954	0.0060	—	1,200
	30	15.30	0.954	0.0060	—	1,200
	30	37.12	0.954	0.0060	—	1,200
<i>$d_z = 1$ clustered DGP (cluster-robust AR)</i>						
	1	8.50	0.958	0.0063	0.809	1,000
	1	5.79	0.953	0.0067	0.667	1,000
<i>$d_z = 1$ survey-weighted DGP (weighted AR of τ_w)</i>						
	1	44.09	0.959	0.0063	—	1,000
	1	156.70	0.960	0.0062	—	1,000

Table B.3: Identification-aware selector benchmark. For each regime we compare the MSE-optimal rule against the concentration (identification-aware) rule, both with correctly specified nuisances. Columns report median effective- F , bias, RMSE, mean Wald CI length, and AR coverage.

Regime	Rule	median eff.- F	bias	RMSE	Wald CI len.	AR cov.
A (weak)	MSE	4.31	-0.0430	2.5276	45.1930	0.932
	concentration	4.73	-0.0962	1.0211	4.6306	0.928
B (moderate)	MSE	11.92	-0.0273	0.3390	1.3231	0.932
	concentration	12.61	-0.0094	0.3335	1.2756	0.928
C (do-no-harm (low absorption))	MSE	23.32	-0.0130	0.2191	0.8711	0.924
	concentration	23.93	-0.0058	0.2145	0.8522	0.924

Table B.4: Card (1995) control-richness ladder (instrument: nearc4 ($d_z = 1$), $N = 2,182$). As controls accumulate, the linear and RF first-stage effective- F collapse and the AR set switches from bounded to the whole line, even though the out-of-fold partial- R^2 and $\text{Var}(\tilde{Z})$ are nearly flat.

Rung	eff.- F (lin. π)	eff.- F (RF π)	f.s. slope	$\text{Var}(\tilde{Z})$	R^2_{of}	AR set
lean	7.06	6.16	0.2551	0.1655	0.23	bounded
+family	4.03	3.67	0.1885	0.1656	0.23	whole line
+parentEd	3.14	3.23	0.1581	0.1659	0.23	whole line
+KWW	4.29	2.68	0.1588	0.1661	0.23	whole line
+poly2	2.92	1.59	0.1399	0.1730	0.20	whole line

Table B.5: OHIE Lee (2009) trimming bounds on the treatment effect under differential survey response. We report the trimming fraction, treatment/control response rates, the first-stage compliance, and the resulting ITT and LATE bounds.

Quantity	Value
Trimming fraction q	0.331
Response rate, treated	0.396
Response rate, control	0.265
First-stage compliance	0.294
ITT point estimate	0.249
ITT Lee bound $[_{-}, -]$	$[-1.1663, 1.2767]$
LATE point estimate	0.846
LATE Lee bound $[_{-}, -]$	$[-3.961, 4.336]$

Table B.6: Angrist–Krueger (1991) over-identified inference. Top panel: OLS, 2SLS, and DML point estimates with weak-IV-robust AR/LM/CLR sets and the Hansen- J over-identification test for each instrument block. Bottom panel: Monte-Carlo AR/LM/CLR coverage and Hansen- J rejection (size) at $d_z = 3, 30$ from the over-identified coverage certificate.

Spec.	Est.	$\hat{\tau}$	SE	eff.- F	AR set	LM set	CLR set	Hansen J	J p
<i>Quarter of birth ($d_z = 3$), $N = 329,509$, $\text{df}_J = 2$</i>									
	OLS	0.0673	0.0013	—	—	—	—	2.84	0.2421
	2SLS	0.1077	0.0170	11.43	—	—	—		
	DML (linear)	0.1077	0.0196	11.43	[0.051, 0.148]	[0.057, 0.139]	[0.058, 0.138]		
<i>QOB $\times$ YOY ($d_z = 30$), $N = 329,509$, $\text{df}_J = 29$</i>									
	OLS	0.0673	0.0013	—	—	—	—	25.15	0.6705
	2SLS	0.0869	0.0108	12.29	—	—	—		
	DML (linear)	0.0870	0.0159	12.29	whole line	whole line	whole line		

d_z	median eff.- F	AR cov.	LM cov.	CLR cov.	AR MCSE	LM MCSE	CLR MCSE
3	4.23	0.958	0.956	0.953	0.0058	0.0059	0.0061
3	14.82	0.958	0.958	0.955	0.0058	0.0058	0.0060
3	36.49	0.958	0.957	0.956	0.0058	0.0059	0.0059
30	4.58	0.954	0.957	0.955	0.0060	0.0059	0.0060
30	15.30	0.954	0.962	0.960	0.0060	0.0055	0.0057
30	37.12	0.954	0.958	0.958	0.0060	0.0058	0.0058

<i>Hansen-J rejection (size, nominal 0.05)</i>							
$d_z = 3$ (df = 2): rejection = 0.044 (MCSE 0.0092), reps = 500							
$d_z = 30$ (df = 29): rejection = 0.044 (MCSE 0.0092), reps = 500							

Data and code availability

The replication materials reproduce every number, table, and figure in the paper from the committed result artifacts. Deterministic cells (Card, CollegeDistance, Angrist–Krueger) and all Monte-Carlo certificates reproduce bit-for-bit; the examiner cell uses random-forest folds and reproduces up to fold-level sampling. Three public datasets are committed with the code (**CollegeDistance**, Card, Angrist–Krueger); two are governed by their providers’ access terms and must be obtained from source (the Oregon Health Insurance Experiment from NBER and the Miami–Dade examiner data from openICPSR 209883), with placement instructions in the accompanying **REPRODUCE.md**. The first-stage, coverage-certificate, selector, and over-identified analyses are produced by the scripts in **Reproduction code/** and orchestrated by **run_all.py**; all figures are regenerated from the committed JSON artifacts by **make_figures.py**.

References

- Amemiya, T. (1974). The nonlinear two-stage least-squares estimator. *Journal of Econometrics* 2(2): 105–110.
- Anderson, T. W., and H. Rubin (1949). Estimation of the parameters of a single equation in a complete system of stochastic equations. *Annals of Mathematical Statistics* 20(1): 46–63.
- Andrews, I., J. H. Stock, and L. Sun (2019). Weak instruments in instrumental variables regression: Theory and practice. *Annual Review of Economics* 11: 727–753.

- Angrist, J. D., and B. Frandsen (2022). Machine labor. *Journal of Labor Economics* 40(S1): S97–S140.
- Angrist, J. D., G. W. Imbens, and A. B. Krueger (1999). Jackknife instrumental variables estimation. *Journal of Applied Econometrics* 14(1): 57–67.
- Angrist, J. D., and A. B. Krueger (1991). Does compulsory school attendance affect schooling and earnings? *Quarterly Journal of Economics* 106(4): 979–1014.
- Athey, S., J. Tibshirani, and S. Wager (2019). Generalized random forests. *Annals of Statistics* 47(2): 1148–1178.
- Belloni, A., D. Chen, V. Chernozhukov, and C. Hansen (2012). Sparse models and methods for optimal instruments with an application to eminent domain. *Econometrica* 80(6): 2369–2429.
- Blandhol, C., J. Bonney, M. Mogstad, and A. Torgovitsky (2026). When is TSLS actually LATE? *Review of Economic Studies*, forthcoming (doi:10.1093/restud/rdag029).
- Bound, J., D. A. Jaeger, and R. M. Baker (1995). Problems with instrumental variables estimation when the correlation between the instruments and the endogenous explanatory variable is weak. *Journal of the American Statistical Association* 90(430): 443–450.
- Card, D. (1995). Using geographic variation in college proximity to estimate the return to schooling. In L. N. Christofides, E. K. Grant, and R. Swidinsky (eds.), *Aspects of Labour Market Behaviour: Essays in Honour of John Vanderkamp*, 201–222. University of Toronto Press.
- Çetin, T. (2026). Nested Cross-Validated Double Machine Learning for the Partially Linear IV Model. Manuscript submitted to *The Econometrics Journal*.
- Chamberlain, G. (1987). Asymptotic efficiency in estimation with conditional moment restrictions. *Journal of Econometrics* 34(3): 305–334.
- Chao, J. C., and N. R. Swanson (2005). Consistent estimation with a large number of weak instruments. *Econometrica* 73(5): 1673–1692.
- Chernozhukov, V., D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins (2018). Double/debiased machine learning for treatment and structural parameters. *Econometrics Journal* 21(1): C1–C68.
- Chernozhukov, V., J. C. Escanciano, H. Ichimura, W. K. Newey, and J. M. Robins (2022). Locally robust semiparametric estimation. *Econometrica* 90(4): 1501–1535.
- Chyn, E., B. Frandsen, and E. C. Leslie (2025). Examiner and judge designs in economics: A practitioner’s guide. *Journal of Economic Literature* 63(2): 401–439.
- Davidson, R., and J. G. MacKinnon (2014). Confidence sets based on inverting Anderson–Rubin tests. *Econometrics Journal* 17(2): S39–S58.
- Dobbie, W., J. Goldin, and C. S. Yang (2018). The effects of pretrial detention on conviction, future crime, and employment: Evidence from randomly assigned judges. *American Economic Review* 108(2): 201–240.
- Finkelstein, A., S. Taubman, B. Wright, M. Bernstein, J. Gruber, J. P. Newhouse, H. Allen, K. Baicker, and the Oregon Health Study Group (2012). The Oregon health insurance experiment: Evidence from the first year. *Quarterly Journal of Economics* 127(3): 1057–1106.
- Frandsen, B., E. Leslie, and S. McIntyre (2025). Cluster jackknife instrumental variables estimation. *Review of Economics and Statistics*, forthcoming (doi:10.1162/rest.a.263).

- Hansen, C., J. Hausman, and W. Newey (2008). Estimation with many instrumental variables. *Journal of Business & Economic Statistics* 26(4): 398–422.
- Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica* 50(4): 1029–1054.
- Hartford, J., G. Lewis, K. Leyton-Brown, and M. Taddy (2017). Deep IV: A flexible approach for counterfactual prediction. *Proceedings of ICML*, 1414–1423.
- Kleibergen, F. (2002). Pivotal statistics for testing structural parameters in instrumental variables regression. *Econometrica* 70(5): 1781–1803.
- Kolesár, M. (2013). Estimation in an instrumental variables model with treatment effect heterogeneity. Working paper, Princeton University.
- Lee, D. S. (2009). Training, wages, and sample selection: Estimating sharp bounds on treatment effects. *Review of Economic Studies* 76(3): 1071–1102.
- Lee, D. S., J. McCrary, M. J. Moreira, and J. Porter (2022). Valid t -ratio inference for IV. *American Economic Review* 112(10): 3260–3290.
- Mikusheva, A., and L. Sun (2022). Inference with many weak instruments. *Review of Economic Studies* 89(5): 2663–2686.
- Montiel Olea, J. L., and C. Pflueger (2013). A robust test for weak instruments. *Journal of Business & Economic Statistics* 31(3): 358–369.
- Moreira, M. J. (2003). A conditional likelihood ratio test for structural models. *Econometrica* 71(4): 1027–1048.
- Newey, W. K. (1990). Efficient instrumental variables estimation of nonlinear models. *Econometrica* 58(4): 809–837.
- Rouse, C. E. (1995). Democratization or diversion? The effect of community colleges on educational attainment. *Journal of Business & Economic Statistics* 13(2): 217–224.
- Sanderson, E., and F. Windmeijer (2016). A weak instrument F -test in linear IV models with multiple endogenous variables. *Journal of Econometrics* 190(2): 212–221.
- Sargan, J. D. (1958). The estimation of economic relationships using instrumental variables. *Econometrica* 26(3): 393–415.
- Staiger, D., and J. H. Stock (1997). Instrumental variables regression with weak instruments. *Econometrica* 65(3): 557–586.